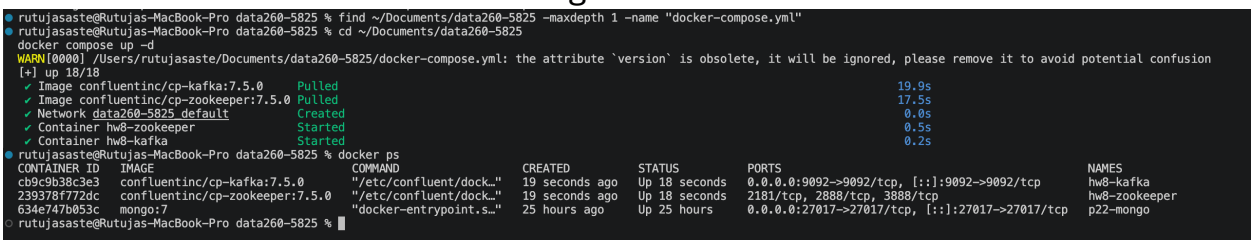
DATA 260
Homework 8 Report

GITHUB REPOSITORY: https://github.com/rutujasaste06/data260-5825
Student ID- 020065825
SID4- 5825
PORT_BASE - 8425
PREFIX - s5825
SEED - 5825
VERIFY_SEED – 265825
DOMAIN_ID - 1
DOMAIN - Clinical trial listings
HARDWARE - MacBook Pro, Apple M5 chip
LOCAL MODEL USED: qwen3:8b (Ollama)
TAGGED COMMIT HASH: 7a638636b79295fdc4c1db12811769b02dd56ac5

Outputs:

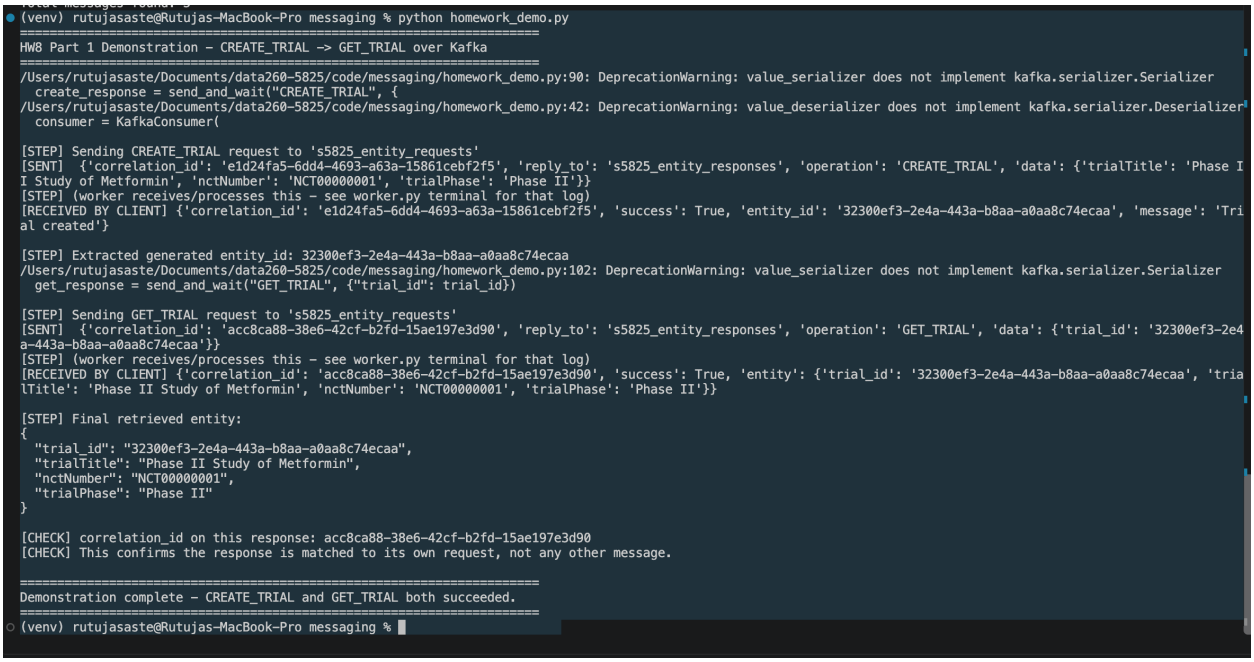
Docker Kafka Infrastructure Running



Part 1 - Backend Service Request/Response over Kafka:

Topics used: `s5825_entity_requests` (client sends requests here) and `s5825_entity_responses` (worker sends responses here).

Kafka Client/Worker -CREATE_Trial and GET_Trial Round Trip:
Client sends a request with a correlation_id over the requests topic; worker processes it and replies on the responses topic; client matches the response to its original request by that same correlation_id.
Client-side log showing correlation_id creation, CREATE_TRIAL send/receive, entity_id extraction, GET_TRIAL send/receive, and final correlation_id confirmation, per the assignment's required sequence.
Every request message contains four fields: correlation_id (a unique ticket number), reply_to (which topic to answer on), operation (CREATE_TRIAL or GET_TRIAL), and data (the trial fields)



```
(venv) rutujasaste@Rutujas-MacBook-Pro messaging % python homework_demo.py
============================================================
HW8 Part 1 Demonstration — CREATE_TRIAL -> GET_TRIAL over Kafka
============================================================
/Users/rutujasaste/Documents/data260-5825/code/messaging/homework_demo.py:90: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  create_response = send_and_wait("CREATE_TRIAL", {
/Users/rutujasaste/Documents/data260-5825/code/messaging/homework_demo.py:42: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(

[STEP] Sending CREATE_TRIAL request to 's5825_entity_requests'
[SENT]  {'correlation_id': 'e1d24fa5-6dd4-4693-a63a-15861cebf2f5', 'reply_to': 's5825_entity_responses', 'operation': 'CREATE_TRIAL', 'data': {'trialTitle': 'Phase I
I Study of Metformin', 'nctNumber': 'NCT00000001', 'trialPhase': 'Phase II'}}
[STEP] (worker receives/processes this — see worker.py terminal for that log)
[RECEIVED BY CLIENT] {'correlation_id': 'e1d24fa5-6dd4-4693-a63a-15861cebf2f5', 'success': True, 'entity_id': '32300ef3-2e4a-443a-b8aa-a0aa8c74ecaa', 'message': 'Tri
al created'}

[STEP] Extracted generated entity_id: 32300ef3-2e4a-443a-b8aa-a0aa8c74ecaa
/Users/rutujasaste/Documents/data260-5825/code/messaging/homework_demo.py:102: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  get_response = send_and_wait("GET_TRIAL", {"trial_id": trial_id})

[STEP] Sending GET_TRIAL request to 's5825_entity_requests'
[SENT]  {'correlation_id': 'acc8ca88-38e6-42cf-b2fd-15ae197e3d90', 'reply_to': 's5825_entity_responses', 'operation': 'GET_TRIAL', 'data': {'trial_id': '32300ef3-2e4
a-443a-b8aa-a0aa8c74ecaa'}}
[STEP] (worker receives/processes this — see worker.py terminal for that log)
[RECEIVED BY CLIENT] {'correlation_id': 'acc8ca88-38e6-42cf-b2fd-15ae197e3d90', 'success': True, 'entity': {'trial_id': '32300ef3-2e4a-443a-b8aa-a0aa8c74ecaa', 'tria
lTitle': 'Phase II Study of Metformin', 'nctNumber': 'NCT00000001', 'trialPhase': 'Phase II'}}

[STEP] Final retrieved entity:
{
  "trial_id": "32300ef3-2e4a-443a-b8aa-a0aa8c74ecaa",
  "trialTitle": "Phase II Study of Metformin",
  "nctNumber": "NCT00000001",
  "trialPhase": "Phase II"
}

[CHECK] correlation_id on this response: acc8ca88-38e6-42cf-b2fd-15ae197e3d90
[CHECK] This confirms the response is matched to its own request, not any other message.

============================================================
Demonstration complete — CREATE_TRIAL and GET_TRIAL both succeeded.
============================================================
(venv) rutujasaste@Rutujas-MacBook-Pro messaging %
```

Every response message echoes back the same correlation_id, plus success (true/false). A successful CREATE_TRIAL also includes the generated entity_id; a successful GET_TRIAL includes the full entity; a failure includes an error message.

Code:

```python
def validate_data(data) -> str | None:
    """Returns an error message if invalid, or None if the data is OK."""
    if not isinstance(data, dict):
        return "data must be an object"
    for field in REQUIRED_FIELDS:
        value = data.get(field)
        if value is None or (isinstance(value, str) and value.strip() == ""):
            return f"Missing or empty required field: {field}"
    return None
```

```python
def process_operation(correlation_id, operation, data) -> dict:
    # permanently broken/corrupted message (not a normal validation error).
    if isinstance(data, dict) and data.get("trialTitle") == "POISON_TEST_TRIGGER_CRASH":
        raise RuntimeError("Simulated processing crash (poison message test)")

    if operation == "CREATE_TRIAL":
        error = validate_data(data)
        if error:
            return {
                "correlation_id": correlation_id,
                "success": False,
                "error": error,
            }
        trial_id = str(uuid.uuid4())
        record = {"trial_id": trial_id, **data}
        trials_db[trial_id] = record
        return {
            "correlation_id": correlation_id,
            "success": True,
            "entity_id": trial_id,
            "message": "Trial created",
        }

    elif operation == "GET_TRIAL":
        trial_id = data.get("trial_id") if isinstance(data, dict) else None
        record = trials_db.get(trial_id)
        if record is None:
            return {
                "correlation_id": correlation_id,
                "success": False,
                "error": "Entity not found",
            }
        return {
            "correlation_id": correlation_id,
            "success": True,
            "entity": record,
        }

    else:
        return {
            "correlation_id": correlation_id,
            "success": False,
            "error": "Unknown operation",
        }
```

```python
try:
    response = process_operation(correlation_id, operation, data)
    processed_requests[correlation_id] = response

    reply_to = request.get("reply_to", RESPONSE_TOPIC)
    producer.send(reply_to, value=response)
    producer.flush()
    print(f"[worker] Sent response to {reply_to}: {response}")
```

```
(venv) rutujasaste@Rutujas-MacBook-Pro messaging %
python worker.py
/Users/rutujasaste/Documents/data260-5825/code/messaging/worker.py:81: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
/Users/rutujasaste/Documents/data260-5825/code/messaging/worker.py:108: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
Consumer is ready and listening...
[worker] Received request: {'correlation_id': 'a6b73bbe-3eaa-4f05-9591-c3e0716d8488', 'reply_to': 's5825_entity_responses', 'operation': 'CREATE_TRIAL', 'data': {'tr
ialTitle': 'Phase II Study of Metformin', 'nctNumber': 'NCT00000001', 'trialPhase': 'Phase II'}}
[worker] Sent response to s5825_entity_responses: {'correlation_id': 'a6b73bbe-3eaa-4f05-9591-c3e0716d8488', 'success': True, 'entity_id': '6e95e21a-5df5-48e4-b804-8
6ede60301fb', 'message': 'Trial created'}
[worker] Received request: {'correlation_id': '56652aa2-6584-41c9-9c3a-3c2a226c390c', 'reply_to': 's5825_entity_responses', 'operation': 'GET_TRIAL', 'data': {'trial
_id': '6e95e21a-5df5-48e4-b804-86ede60301fb'}}
[worker] Sent response to s5825_entity_responses: {'correlation_id': '56652aa2-6584-41c9-9c3a-3c2a226c390c', 'success': True, 'entity': {'trial_id': '6e95e21a-5df5-4
8e4-b804-86ede60301fb', 'trialTitle': 'Phase II Study of Metformin', 'nctNumber': 'NCT00000001', 'trialPhase': 'Phase II'}}
[worker] Received request: {'correlation_id': '0c933d12-fb84-44ea-90a9-9d3eff136baf', 'reply_to': 's5825_entity_responses', 'operation': 'CREATE_TRIAL', 'data': {'tr
ialTitle': 'Phase II Study of Metformin', 'nctNumber': 'NCT00000001', 'trialPhase': 'Phase II'}}
[worker] Sent response to s5825_entity_responses: {'correlation_id': '0c933d12-fb84-44ea-90a9-9d3eff136baf', 'success': True, 'entity_id': '1a90cf8a-8137-42fe-9189-6
13bf9021ff2', 'message': 'Trial created'}
[worker] Received request: {'correlation_id': '055c0f7e-d30b-47c7-98c6-30a395d8c296', 'reply_to': 's5825_entity_responses', 'operation': 'CREATE_TRIAL', 'data': {'tr
ialTitle': 'Phase II Study of Metformin', 'nctNumber': 'NCT00000001', 'trialPhase': 'Phase II'}}
[worker] Sent response to s5825_entity_responses: {'correlation_id': '055c0f7e-d30b-47c7-98c6-30a395d8c296', 'success': True, 'entity_id': '2afc1763-677f-4a4f-8f03-0
4570022f5fa', 'message': 'Trial created'}
[worker] Received request: {'correlation_id': '15a2a7f1-0961-40ab-911a-b2ccbe3425b9', 'reply_to': 's5825_entity_responses', 'operation': 'CREATE_TRIAL', 'data': {'tr
ialTitle': 'Phase II Study of Metformin', 'nctNumber': 'NCT00000001', 'trialPhase': 'Phase II'}}
[worker] Sent response to s5825_entity_responses: {'correlation_id': '15a2a7f1-0961-40ab-911a-b2ccbe3425b9', 'success': True, 'entity_id': 'b5170d73-4258-4bf5-8d11-4
fd881b41da6', 'message': 'Trial created'}
[worker] Received request: {'correlation_id': 'e1d24fa5-6dd4-4693-a63a-15861cebf2f5', 'reply_to': 's5825_entity_responses', 'operation': 'CREATE_TRIAL', 'data': {'tr
ialTitle': 'Phase II Study of Metformin', 'nctNumber': 'NCT00000001', 'trialPhase': 'Phase II'}}
[worker] Sent response to s5825_entity_responses: {'correlation_id': 'e1d24fa5-6dd4-4693-a63a-15861cebf2f5', 'success': True, 'entity_id': '32300ef3-2e4a-443a-b8aa-a
0aa8c74ecaa', 'message': 'Trial created'}
[worker] Received request: {'correlation_id': 'acc8ca88-38e6-42cf-b2fd-15ae197e3d90', 'reply_to': 's5825_entity_responses', 'operation': 'GET_TRIAL', 'data': {'trial
_id': '32300ef3-2e4a-443a-b8aa-a0aa8c74ecaa'}}
[worker] Sent response to s5825_entity_responses: {'correlation_id': 'acc8ca88-38e6-42cf-b2fd-15ae197e3d90', 'success': True, 'entity': {'trial_id': '32300ef3-2e4a-4
43a-b8aa-a0aa8c74ecaa', 'trialTitle': 'Phase II Study of Metformin', 'nctNumber': 'NCT00000001', 'trialPhase': 'Phase II'}}
```

The client generates a new correlation_id for each request; the worker copies that same correlation_id unchanged into its response, so the client can match each reply back to the exact request that caused it, even if multiple requests are in flight at once.

Required 10-Step Demonstration:

```
(venv) rutujasaste@Rutujas-MacBook-Pro messaging %
python worker.py
/Users/rutujasaste/Documents/data260-5825/code/messaging/worker.py:81: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
/Users/rutujasaste/Documents/data260-5825/code/messaging/worker.py:108: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
Consumer is ready and listening...


(venv) rutujasaste@Rutujas-MacBook-Pro messaging % python client.py
/Users/rutujasaste/Documents/data260-5825/code/messaging/client.py:67: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  created = send_request("CREATE_TRIAL", {
/Users/rutujasaste/Documents/data260-5825/code/messaging/client.py:27: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
[client] Sending CREATE_TRIAL (correlation_id=a6b73bbe-3eaa-4f05-9591-c3e0716d8488): {'correlation_id': 'a6b73bbe-3eaa-4f05-9591-c3e0716d8488', 'reply_to': 's5825_en
tity_responses', 'operation': 'CREATE_TRIAL', 'data': {'trialTitle': 'Phase II Study of Metformin', 'nctNumber': 'NCT00000001', 'trialPhase': 'Phase II'}}
[client] Received response: {'correlation_id': 'a6b73bbe-3eaa-4f05-9591-c3e0716d8488', 'success': True, 'entity_id': '6e95e21a-5df5-48e4-b804-86ede60301fb', 'message
': 'Trial created'}
/Users/rutujasaste/Documents/data260-5825/code/messaging/client.py:74: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  fetched = send_request("GET_TRIAL", {"trial_id": trial_id})
[client] Sending GET_TRIAL (correlation_id=56652aa2-6584-41c9-9c3a-3c2a226c390c): {'correlation_id': '56652aa2-6584-41c9-9c3a-3c2a226c390c', 'reply_to': 's5825_entit
y_responses', 'operation': 'GET_TRIAL', 'data': {'trial_id': '6e95e21a-5df5-48e4-b804-86ede60301fb'}}
[client] Received response: {'correlation_id': '56652aa2-6584-41c9-9c3a-3c2a226c390c', 'success': True, 'entity': {'trial_id': '6e95e21a-5df5-48e4-b804-86ede60301fb'
, 'trialTitle': 'Phase II Study of Metformin', 'nctNumber': 'NCT00000001', 'trialPhase': 'Phase II'}}
(venv) rutujasaste@Rutujas-MacBook-Pro messaging %
```

Code:

```python
def send_and_wait(operation: str, data: dict, timeout_seconds: int = 15) -> dict:
    # Step 1: create one correlation_id for this request
    correlation_id = str(uuid.uuid4())

    producer = KafkaProducer(
        bootstrap_servers=KAFKA_BROKER,
        value_serializer=lambda v: json.dumps(v).encode("utf-8"),
    )

    consumer = KafkaConsumer(
        bootstrap_servers=KAFKA_BROKER,
        value_deserializer=lambda v: json.loads(v.decode("utf-8")),
    )
    partitions = consumer.partitions_for_topic(RESPONSE_TOPIC) or {0}
    topic_partitions = [TopicPartition(RESPONSE_TOPIC, p) for p in partitions]
    consumer.assign(topic_partitions)
    consumer.seek_to_beginning(*topic_partitions)  # scan all responses, filter by correlation_id below

    request = {
        "correlation_id": correlation_id,
        "reply_to": RESPONSE_TOPIC,
        "operation": operation,
        "data": data,
    }

    # Step 2 + 3: send the request, log what was sent
    print(f"\n[STEP] Sending {operation} request to '{REQUEST_TOPIC}'")
    print(f"[SENT]  {request}")
    producer.send(REQUEST_TOPIC, value=request)
        for _tp, messages in records.items():
            for msg in messages:
                response = msg.value
                if response.get("correlation_id") == correlation_id:
                    print(f"[RECEIVED BY CLIENT] {response}")
                    consumer.close()
                    producer.close()
                    return response

    consumer.close()
    producer.close()
    raise TimeoutError(f"No response received for correlation_id {correlation_id} within {timeout_seconds}s")
# Step 10: display the retrieved entity and confirm correlation_id preserved
assert get_response.get("success") is True, "GET_TRIAL did not succeed"
print("\n[STEP] Final retrieved entity:")
print(json.dumps(get_response["entity"], indent=2))
print(f"\n[CHECK] correlation_id on this response: {get_response['correlation_id']}")
```

Docker Containers Running - Kafka, Worker, and FastAPI

All required services running as Docker containers: Kafka broker, the Kafka worker (consumer/producer), and the FastAPI service, confirmed via docker ps and their startup logs.

```
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % docker ps
CONTAINER ID   IMAGE                            COMMAND                  CREATED          STATUS          PORTS                                                  NAMES
462775347ed3   data260-5825-api                 "uvicorn api:app --h…"   57 seconds ago   Up 57 seconds   0.0.0.0:8425->8425/tcp, [::]:8425->8425/tcp            hw8-api
0a4f888258d2   data260-5825-worker              "python worker.py"       57 seconds ago   Up 57 seconds                                                          hw8-work
er
78402204fd26   confluentinc/cp-kafka:7.5.0      "/etc/confluent/dock…"   58 seconds ago   Up 57 seconds   0.0.0.0:9092->9092/tcp, [::]:9092->9092/tcp            hw8-kafk
a
239378f772dc   confluentinc/cp-zookeeper:7.5.0  "/etc/confluent/dock…"   2 hours ago      Up 2 hours      2181/tcp, 2888/tcp, 3888/tcp                           hw8-zook
eeper
634e747b053c   mongo:7                          "docker-entrypoint.s…"   27 hours ago     Up 27 hours     0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp        p22-mong
o
```

Consumer is ready and listening:

```
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % docker logs hw8-worker
docker logs hw8-api
  main()
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8425 (Press CTRL+C to quit)
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % docker compose up -d --build worker api
docker logs hw8-worker
docker logs hw8-api
WARN[0000] /Users/rutujasaste/Documents/data260-5825/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential
 confusion
[+] Building 8.6s (14/14) FINISHED
 => [internal] load local bake definitions                                                                                                                0.0s
 => => reading from stdin 1.05kB                                                                                                                          0.0s
 => [api internal] load build definition from Dockerfile                                                                                                  0.0s
 => => transferring dockerfile: 296B                                                                                                                      0.0s
 => [worker internal] load metadata for docker.io/library/python:3.11-slim                                                                                0.7s
 => [worker internal] load .dockerignore                                                                                                                  0.0s
 => => transferring context: 2B                                                                                                                           0.0s
 => [worker internal] load build context                                                                                                                  0.0s
 => => transferring context: 565B                                                                                                                         0.0s
 => [api 1/5] FROM docker.io/library/python:3.11-slim@sha256:673719e071ecf27434fd2635bbd8d00f49f4fdd81f243143438a42ced8bedf60                             0.0s
 => => resolve docker.io/library/python:3.11-slim@sha256:673719e071ecf27434fd2635bbd8d00f49f4fdd81f243143438a42ced8bedf60                                 0.0s
 => CACHED [api 2/5] WORKDIR /app                                                                                                                         0.0s
 => CACHED [api 3/5] COPY requirements.txt .                                                                                                              0.0s
 => [worker 4/5] RUN pip install --no-cache-dir -r requirements.txt                                                                                       6.8s
 => [api 5/5] COPY . .                                                                                                                                    0.0s
 => [api] exporting to image                                                                                                                              0.8s
 => => exporting layers                                                                                                                                   0.5s
 => => exporting manifest sha256:2cb0a89da8936c3e757c33de46a201b5bb256d07d5e1d356e1160f7d61d0fc22                                                          0.0s
 => => exporting config sha256:d78b45ef640a7e2f2c76b74b9a6626c02f9f8b27798b5471a5e042959d4f92e3                                                           0.0s
 => => exporting attestation manifest sha256:6dcba234aea7e0b0fae79bed0be98ea137a8132bde22ef1ebd4a2f5a2c21988c                                             0.0s
 => => exporting manifest list sha256:6bd87b223be480d6da336b31644bfeeff63937276e1e806715237ce0cc35bc16                                                    0.0s
 => => naming to docker.io/library/data260-5825-api:latest                                                                                                0.0s
 => => unpacking to docker.io/library/data260-5825-api:latest                                                                                             0.2s
 => [worker] exporting to image                                                                                                                           0.8s
 => => exporting layers                                                                                                                                   0.5s
 => => exporting manifest sha256:14c7c0a6c46ff9d2cf8c4e6a0dfcba869582cf557910d91972e691c8b81d89b4                                                          0.0s
 => => exporting config sha256:b79b8537d025d8c8b3e4c0a3b17ab1f431a89321167e2f535ed84dd388d4ed48                                                           0.0s
 => => exporting attestation manifest sha256:bebc6fe64a11a50c3ca39aa1ddc0682d829afbab13b59d29a2ce974fdb585135                                             0.0s
 => => exporting manifest list sha256:8d7f5398aea7aa337001b28d45344a3beff54f6b4bda159b44b50df1b377d43d                                                    0.0s
 => => naming to docker.io/library/data260-5825-worker:latest                                                                                             0.0s
 => => unpacking to docker.io/library/data260-5825-worker:latest                                                                                          0.2s
 => [worker] resolving provenance for metadata file                                                                                                       0.0s
 => [api] resolving provenance for metadata file                                                                                                          0.0s
[+] up 6/6
 ✔ Image data260-5825-worker Built                                                                                                                        8.7s
 ✔ Image data260-5825-api    Built                                                                                                                        8.7s
 ✔ Container hw8-zookeeper   Running                                                                                                                      0.0s
 ✔ Container hw8-kafka       Running                                                                                                                      0.0s
 ✔ Container hw8-worker      Started                                                                                                                      3.6s
 ✔ Container hw8-api         Started                                                                                                                      0.4s
/app/worker.py:83: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
/app/worker.py:110: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
Consumer is ready and listening...
```

```
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % docker logs hw8-api
INFO:     Started server process [1]
INFO:     Waiting for application startup.
FastAPI server started at http://127.0.0.1:8425
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8425 (Press CTRL+C to quit)
```

FastAPI Endpoints : CREATE_TRIAL and GET_TRIAL via Kafka

Trial created

```
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % curl -X POST http://localhost:8425/trials -H "Content-Type: application/json" -d '{"trialTitle": "Phase III Stu
dy of Aspirin", "nctNumber": "NCT00000002", "trialPhase": "Phase III"}'
{"correlation_id":"897bf5f1-e8bb-4b51-937c-a6a839ca86e0","success":true,"entity_id":"1aedaa08-3304-4a37-9fba-2eac6615406a","message":"Trial created"}%
```

GET Trial

```
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % curl http://localhost:8425/trials/1aedaa08-3304-4a37-9fba-2eac6615406a
{"correlation_id":"f410e56d-fa64-4893-9b0a-9766917e93b7","success":true,"entity":{"trial_id":"1aedaa08-3304-4a37-9fba-2eac6615406a","trialTitle":"Phase III Study of
Aspirin","nctNumber":"NCT00000002","trialPhase":"Phase III"}}%
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 %
```

Swagger outputs:

CREATE Trial:

HW8 Kafka Trial API 0.1.0 OAS 3.1

/openapi.json

default

POST /trials Create Trial

Parameters | Cancel | Reset

No parameters

Request body required | application/json

Edit Value | Schema

```
{
  "trialTitle": "Phase I Study of Ibuprofen",
  "nctNumber": "NCT00000003",
  "trialPhase": "Phase I"
}
```

Execute | Clear

O/P: The entity ID is returned:

localhost:8425/docs#/default/create_trial_trials_post

Relaunch to update

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8425/trials' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
  "trialTitle": "Phase I Study of Ibuprofen",
  "nctNumber": "NCT00000003",
  "trialPhase": "Phase I"
}'
```

Request URL

```
http://localhost:8425/trials
```

Server response

Code	Details
200	Response body

```
{
  "correlation_id": "f01d4c95-7c99-44ff-bdea-7265deea71db",
  "success": true,
  "entity_id": "f113c59b-5749-4c74-ba02-ff237bbc9229",
  "message": "Trial created"
}
```

Download

Response headers

```
access-control-allow-credentials: true
access-control-allow-origin: http://localhost:8425
content-length: 149
content-type: application/json
date: Sat,31 Oct 2026 06:49:12 GMT
server: uvicorn
vary: Origin
```

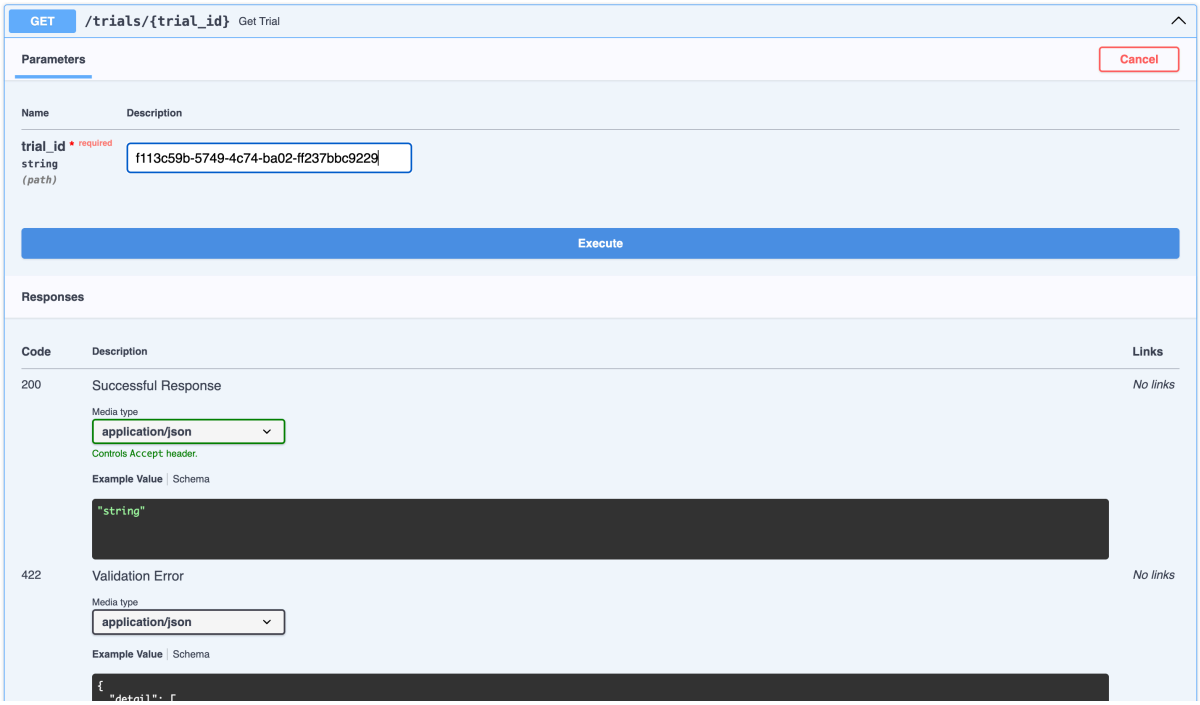
Responses

Code	Description	Links
200	Successful Response	*No links*

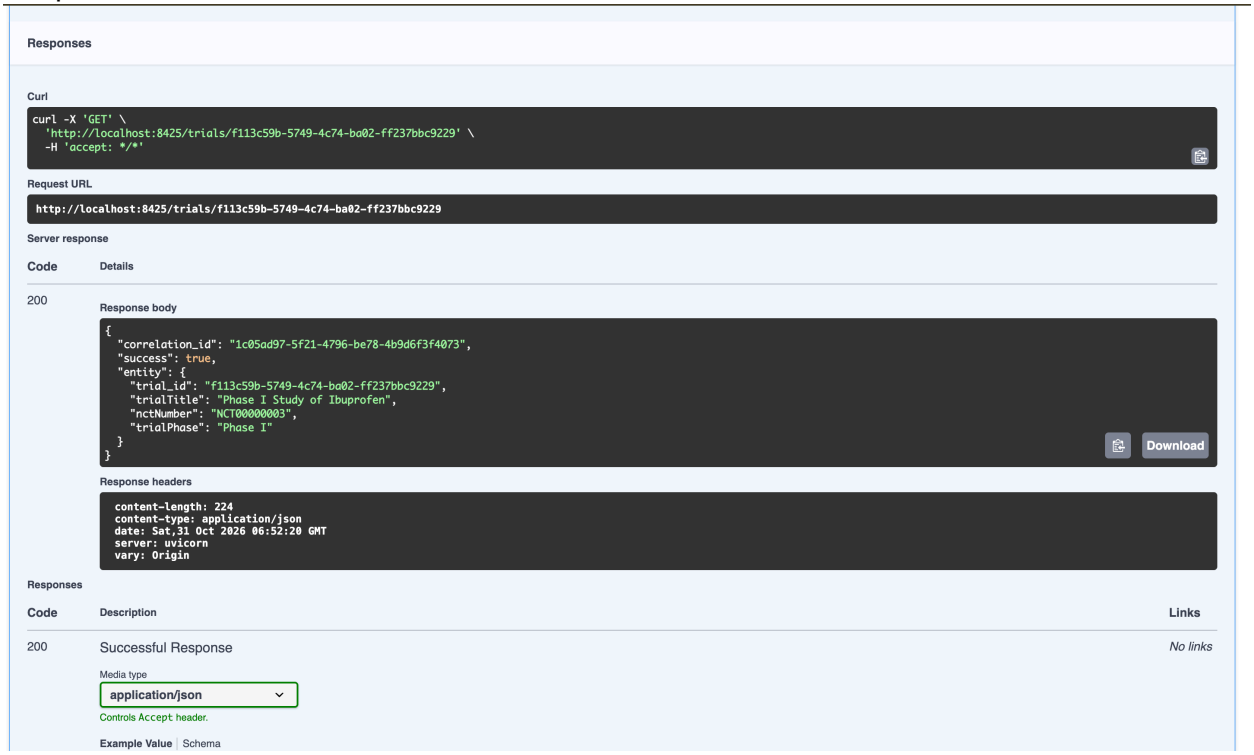
Media type: application/json

Controls Accept header.

GET_TRIAL: Entered the entity id as trial id



Response:



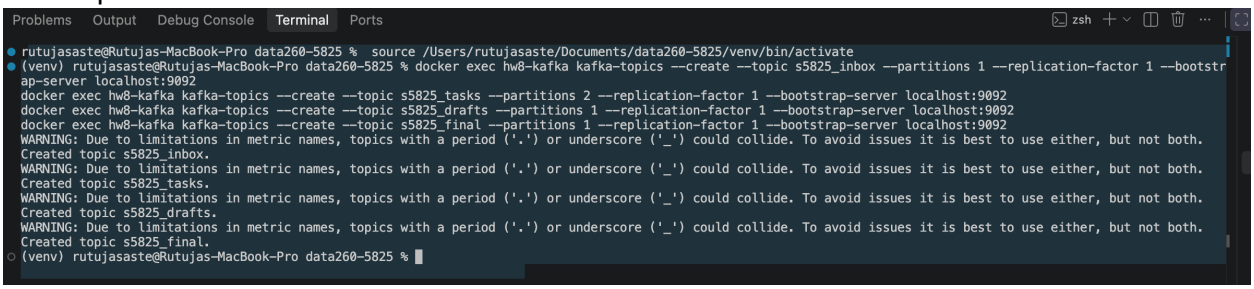
Part 2 - Three-Agent Kafka Pipeline

I. Topics and Workflow

Sender	Reads Topic	Sends Message To
You	-	s5825_inbox (a question)

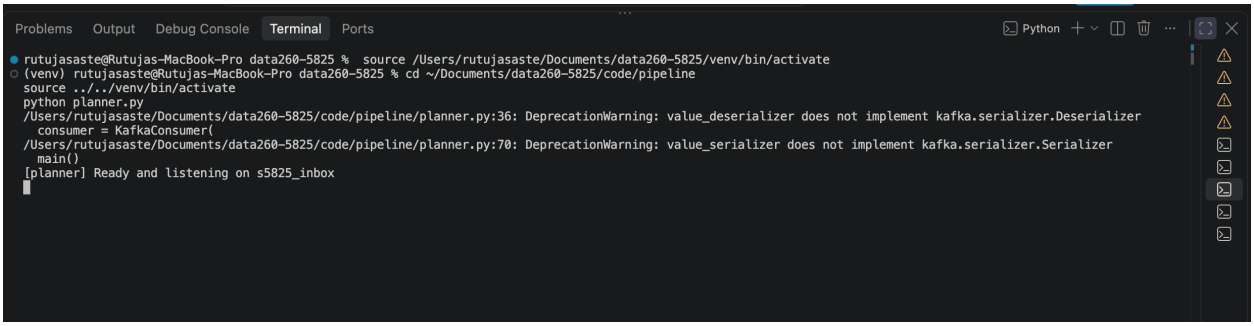
Sender	Reads Topic	Sends Message To
Planner	s5825_inbox	s5825_tasks (a plan)
Writer	s5825_tasks	s5825_drafts (a draft answer)
Reviewer	s5825_drafts	s5825_final (an approval)

All 4 topics created:



Executed

Planner:



Code:



Writer:



Code:

```python
def main():
    consumer = KafkaConsumer(
        TASKS_TOPIC,
        bootstrap_servers=KAFKA_BROKER,
        value_deserializer=lambda v: json.loads(v.decode("utf-8")),
        auto_offset_reset="earliest",
        group_id="s5825-writer-group",
        enable_auto_commit=False,  # we commit manually, only after success - see docstring
        max_poll_records=1,        # hand me ONE message at a time so a commit covers only it
    )
    producer = KafkaProducer(
print(f"[{WRITER_NAME}] Received plan (correlation_id={correlation_id}) "
      f"partition={msg.partition} offset={msg.offset} - starting work "
      f"(offset NOT committed yet - if I die right now, this message is NOT lost)")

draft = write_answer(question, plan)
producer.send(DRAFTS_TOPIC, value=output)
producer.flush()
print(f"[{WRITER_NAME}] Sent draft to {DRAFTS_TOPIC}")

# Only now - after the work is actually done and sent downstream -
# do we tell Kafka this message is handled.
consumer.commit()
print(f"[{WRITER_NAME}] Offset committed for correlation_id={correlation_id} - now safe")
```

Reviewer:



Code:

```python
def review(draft: str) -> str:
    """Returns 'approved' or 'needs_revision'."""
    if not draft or len(draft.strip()) < MIN_LENGTH:
        return "needs_revision"
    return "approved"
```

```python
for msg in consumer:
    draft_msg = msg.value
    correlation_id = draft_msg.get("correlation_id")
    draft = draft_msg.get("draft")
    print(f"[reviewer] Received draft (correlation_id={correlation_id})")

    status = review(draft)
    print(f"[reviewer] Verdict: {status}")

    output = {
        "correlation_id": correlation_id,
        "status": status,
        "answer": draft,
    }
    producer.send(FINAL_TOPIC, value=output)
    producer.flush()
    print(f"[reviewer] Sent final result to {FINAL_TOPIC}")
```

Complete Message Journey: inbox → tasks → drafts → final:

Planner terminal (showing received question + generated plan + sent)

```
rutujasaste@Rutujas-MacBook-Pro data260-5825 %  source /Users/rutujasaste/Documents/data260-5825/venv/bin/activate
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % cd ~/Documents/data260-5825/code/pipeline
source ../../venv/bin/activate
python planner.py
/Users/rutujasaste/Documents/data260-5825/code/pipeline/planner.py:36: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
/Users/rutujasaste/Documents/data260-5825/code/pipeline/planner.py:70: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
[planner] Ready and listening on s5825_inbox
[planner] Received question (correlation_id=5cdb3983-6975-437d-a3e5-bc5477f081d4): What should a patient know before joining a Phase II clinical trial?
[planner] Generated plan: - Understand the trial's purpose, design, and how it differs from earlier phases (e.g., safety vs. efficacy focus).
- Review informed consent documents, including risks, benefits, procedures, and rights to withdraw.
- Clarify treatment details (e.g., dosage, frequency, duration) and how it compares to standard care.
- Discuss potential side effects, monitoring protocols, and support available during/after the trial.
[planner] Sent plan to s5825_tasks
```

Writer terminal (showing received plan + generated draft + sent):

```
rutujasaste@Rutujas-MacBook-Pro data260-5825 %  source /Users/rutujasaste/Documents/data260-5825/venv/bin/activate
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % cd ~/Documents/data260-5825/code/pipeline
source ../../venv/bin/activate
python writer.py
/Users/rutujasaste/Documents/data260-5825/code/pipeline/writer.py:36: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
/Users/rutujasaste/Documents/data260-5825/code/pipeline/writer.py:71: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
[writer] Ready and listening on s5825_tasks
[writer] Received plan (correlation_id=5cdb3983-6975-437d-a3e5-bc5477f081d4)
[writer] Generated draft: Before joining a Phase II trial, a patient should understand the trial's purpose: evaluating treatment efficacy and monitoring side effects
, unlike Phase I's safety focus. Review the informed consent document to grasp risks, benefits, procedures, and the right to withdraw at any time. Clarify treatment
specifics, such as dosage, frequency, duration, and how it compares to standard care. Discuss potential side effects, monitoring protocols (e.g., regular check-ups,
lab tests), and available support during and after the trial. Ensure clarity on time commitments and how the trial aligns with their health goals. Always ask questio
ns and seek clarification to make an informed decision.
[writer] Sent draft to s5825_drafts
```

Reviewer terminal (showing received draft + verdict: approved + sent)

```
rutujasaste@Rutujas-MacBook-Pro data260-5825 %  source /Users/rutujasaste/Documents/data260-5825/venv/bin/activate
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % cd ~/Documents/data260-5825/code/pipeline
source ../../venv/bin/activate
python reviewer.py
/Users/rutujasaste/Documents/data260-5825/code/pipeline/reviewer.py:28: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
/Users/rutujasaste/Documents/data260-5825/code/pipeline/reviewer.py:62: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
[reviewer] Ready and listening on s5825_drafts
[reviewer] Received draft (correlation_id=5cdb3983-6975-437d-a3e5-bc5477f081d4)
[reviewer] Verdict: approved
[reviewer] Sent final result to s5825_final
```

send_question.py terminal (showing the question sent + the final JSON result):

One full pipeline run showing correlation_id 5cdb3983-6975-437d-a3e5-bc5477f081d4 preserved through every stage — Planner, Writer, and Reviewer each communicating only through Kafka, never directly. Final result: status “approved” with the complete answer.



How Kafka helped your agents coordinate?

Kafka works like a shared message system between the three agents. The agents do not call each other directly; instead, one agent sends a message to a Kafka topic, and the next agent reads it and sends its result to another topic. Each agent can run independently because it only needs to know which topic to read from and where to send its output. The correlation_id acts like a tracking number that stays with the message, helping each agent connect the work to the correct original request, even when many requests are processed at the same time. If one agent is slow or temporarily stops, Kafka keeps the messages safely until that agent is ready to read them.

Part 3 - Failure Behavior:

3.1. Duplicate delivery:

Before fix: Bug demonstration-

Duplicate trials were created when same request with the same correlation id was sent twice.



```python
# Idempotency store: correlation_id -> the response we already sent for it.
processed_requests = {}
# Duplicate delivery check (Part 3.I)
if correlation_id in processed_requests:
    response = processed_requests[correlation_id]
    print(f"[worker] DUPLICATE detected for correlation_id={correlation_id} "
          f"- returning cached response, no new trial created")
    reply_to = request.get("reply_to", RESPONSE_TOPIC)
    producer.send(reply_to, value=response)
    producer.flush()
    print(f"[worker] Sent response to {reply_to}: {response}")
    continue
processed_requests[correlation_id] = response
```

After fixing:
After fixing the code the worker container was rebuilt docker compose up – d – build worker. So after rebuilt docker picked up the updated file.
Checked the logs:

```
(venv) rutujasaste@Rutujas-MacBook-Pro part3 % docker compose up -d --build worker
 WARN[0000] /Users/rutujasaste/Documents/data260-5825/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential
  confusion
 [+] Building 1.0s (12/12) FINISHED
  => [internal] load local bake definitions                                                                                                0.0s
  => => reading from stdin 574B                                                                                                            0.0s
  => [internal] load build definition from Dockerfile                                                                                      0.0s
  => => transferring dockerfile: 296B                                                                                                      0.0s
  => [internal] load metadata for docker.io/library/python:3.11-slim                                                                       0.7s
  => [internal] load .dockerignore                                                                                                         0.0s
  => => transferring context: 2B                                                                                                           0.0s
  => [internal] load build context                                                                                                         0.0s
  => => transferring context: 8.50kB                                                                                                       0.0s
  => [1/5] FROM docker.io/library/python:3.11-slim@sha256:673719e071ecf27434fd2635bbd8d00f49f4fdd81f243143438a42ced8bedf60                 0.0s
  => => resolve docker.io/library/python:3.11-slim@sha256:673719e071ecf27434fd2635bbd8d00f49f4fdd81f243143438a42ced8bedf60                 0.0s
  => CACHED [2/5] WORKDIR /app                                                                                                             0.0s
  => CACHED [3/5] COPY requirements.txt .                                                                                                  0.0s
  => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                                       0.0s
  => [5/5] COPY . .                                                                                                                        0.0s
  => exporting to image                                                                                                                    0.0s
  => => exporting layers                                                                                                                   0.0s
  => => exporting manifest sha256:6e9cca740effdda7e9905417ee1a548380a9bd4613cbf810b59e7ad35757d4bc                                         0.0s
  => => exporting config sha256:36c687800d2090beed1a1a069e8aee763d7c7b4982e90d1840a3581c2039a89d4                                         0.0s
  => => exporting attestation manifest sha256:26559c34545f1784ba38cbb28635fad8390e8a1fec2d98281f15d516112178f9                            0.0s
  => => exporting manifest list sha256:85bbef8e8b8206ebbfbe0a6d57d74fc926eeab23666577092c2b81b589a60563                                   0.0s
  => => naming to docker.io/library/data260-5825-worker:latest                                                                             0.0s
  => => unpacking to docker.io/library/data260-5825-worker:latest                                                                          0.0s
  => resolving provenance for metadata file                                                                                                0.0s
 [+] up 4/4
  ✔ Image data260-5825-worker Built                                                                                                        1.0s
  ✔ Container hw8-zookeeper   Running                                                                                                      0.0s
  ✔ Container hw8-kafka       Running                                                                                                      0.0s
  ✔ Container hw8-worker      Started                                                                                                      3.2s
(venv) rutujasaste@Rutujas-MacBook-Pro part3 % docker logs hw8-worker
/app/worker.py:109: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
/app/worker.py:136: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
Consumer is ready and listening...
```

The same correlation_id is sent twice and it now returns the identical entity_id both times, the worker detects the replay and skips redoing the work.

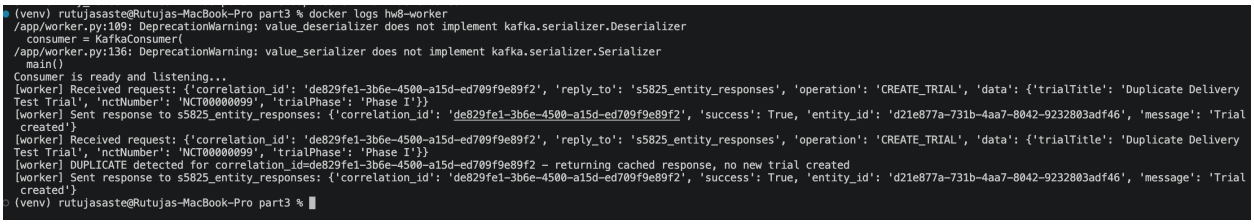
```
(venv) rutujasaste@Rutujas-MacBook-Pro part3 % python test_duplicate_delivery.py
/Users/rutujasaste/Documents/data260-5825/code/messaging/part3/test_duplicate_delivery.py:99: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
/Users/rutujasaste/Documents/data260-5825/code/messaging/part3/test_duplicate_delivery.py:48: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
[test] correlation_id for this test: de829fe1-3b6e-4500-a15d-ed709f9e89f2

[test] Sending CREATE_TRIAL request (1st time, original)
[test] (scanned 17 messages on this topic before finding the match)
[test] Response #1: {'correlation_id': 'de829fe1-3b6e-4500-a15d-ed709f9e89f2', 'success': True, 'entity_id': 'd21e877a-731b-4aa7-8042-9232803adf46', 'message': 'Trial created'}

[test] Replaying the EXACT SAME message (same correlation_id) - simulating a duplicate delivery
[test] (scanned 1 messages on this topic before finding the match)
[test] Response #2: {'correlation_id': 'de829fe1-3b6e-4500-a15d-ed709f9e89f2', 'success': True, 'entity_id': 'd21e877a-731b-4aa7-8042-9232803adf46', 'message': 'Trial created'}

=== RESULT ===
entity_id from response #1: d21e877a-731b-4aa7-8042-9232803adf46
entity_id from response #2: d21e877a-731b-4aa7-8042-9232803adf46
SAME entity_id both times -> idempotent. No duplicate trial was created.
```

The worker also logged the duplicate detection:



3.2. Consumer Death:

Writer was terminated with Ctrl+C while still waiting for the LLM's response, this was done before the draft was generated. Because offset commits happen only after success, Kafka still considers this message unprocessed.



Client Times Out While Writer Is Down:

send_question.py waited 120 seconds but got no final result, because Writer was killed before it could produce the draft. This is the consumer death consequence happened before recovery.

```
rutujasaste@Rutujas-MacBook-Pro data260-5825 %  source /Users/rutujasaste/Documents/data260-5825/venv/bin/activate
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % cd ~/Documents/data260-5825/code/pipeline
source ../../venv/bin/activate
python send_question.py
/Users/rutujasaste/Documents/data260-5825/code/pipeline/send_question.py:67: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
/Users/rutujasaste/Documents/data260-5825/code/pipeline/send_question.py:32: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
[send_question] Sending question to s5825_inbox (correlation_id=05038906-8b41-4514-bf2d-6418f0c6d409)
[send_question] Question: What should a patient know before joining a Phase II clinical trial?
[send_question] Waiting on s5825_final for the matching result ...
Traceback (most recent call last):
  File "/Users/rutujasaste/Documents/data260-5825/code/pipeline/send_question.py", line 67, in <module>
    main()
  File "/Users/rutujasaste/Documents/data260-5825/code/pipeline/send_question.py", line 63, in main
    raise TimeoutError(f"No final result received for correlation_id {correlation_id} within {TIMEOUT_SECONDS}s")
TimeoutError: No final result received for correlation_id 05038906-8b41-4514-bf2d-6418f0c6d409 within 120s
(venv) rutujasaste@Rutujas-MacBook-Pro pipeline %
```

After restarting Writer, the message is redelivered or completed, and validated the offset is committed only after successful processing. The work is not lost and same correlation id was redelivered .

```
(venv) rutujasaste@Rutujas-MacBook-Pro pipeline % python writer.py
/Users/rutujasaste/Documents/data260-5825/code/pipeline/writer.py:45: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
/Users/rutujasaste/Documents/data260-5825/code/pipeline/writer.py:87: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
[writer] Ready and listening on s5825_tasks
[writer] Received plan (correlation_id=05038906-8b41-4514-bf2d-6418f0c6d409) - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer] Generated draft: Before joining a Phase II clinical trial, a patient should understand the trial's purpose: this phase evaluates treatment effectiveness and monitors side effects. They mu
st weigh potential risks, benefits, and how the experimental therapy differs from standard care. Review the informed consent process, which outlines rights to withdraw at any time. Clarify the tri
al's timeline, required visits, and how participation might affect daily life. Ensure all questions are addressed with the research team to make an informed decision. Stay proactive about your hea
lth and treatment goals throughout the process.
[writer] Sent draft to s5825_drafts
[writer] Offset committed for correlation_id=05038906-8b41-4514-bf2d-6418f0c6d409 - now safe
```

Code:

```python
def main():
    consumer = KafkaConsumer(
        TASKS_TOPIC,
        bootstrap_servers=KAFKA_BROKER,
        value_deserializer=lambda v: json.loads(v.decode("utf-8")),
        auto_offset_reset="earliest",
        group_id="s5825-writer-group",
        enable_auto_commit=False,  # we commit manually, only after success - see docstring
        max_poll_records=1,        # hand me ONE message at a time so a commit covers only it
    )
    print(f"[{WRITER_NAME}] Received plan (correlation_id={correlation_id}) "
          f"partition={msg.partition} offset={msg.offset} - starting work "
          f"(offset NOT committed yet - if I die right now, this message is NOT lost)")
draft = write_answer(question, plan)
print(f"[{WRITER_NAME}] Generated draft: {draft}")

output = {
    "correlation_id": correlation_id,
    "question": question,
    "draft": draft,
}
producer.send(DRAFTS_TOPIC, value=output)
producer.flush()
consumer.commit()
print(f"[{WRITER_NAME}] Offset committed for correlation_id={correlation_id} - now safe")
```

Reviewer confirmed it flowed through to the final topic:
The full chain recovered correctly end to end: Writer crashed mid-work -> restarted -> redelivered the same message -> completed it -> Reviewer picked it up -> finalized it.

```
rutujasaste@Rutujas-MacBook-Pro data260-5825 %  source /Users/rutujasaste/Documents/data260-5825/venv/bin/activate
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % cd ~/Documents/data260-5825/code/pipeline
source ../../venv/bin/activate
python reviewer.py
/Users/rutujasaste/Documents/data260-5825/code/pipeline/reviewer.py:28: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
/Users/rutujasaste/Documents/data260-5825/code/pipeline/reviewer.py:62: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
[reviewer] Ready and listening on s5825_drafts
[reviewer] Received draft (correlation_id=5cdb3983-6975-437d-a3e5-bc5477f081d4)
[reviewer] Verdict: approved
[reviewer] Sent final result to s5825_final
[reviewer] Received draft (correlation_id=05038906-8b41-4514-bf2d-6418f0c6d409)
[reviewer] Verdict: approved
[reviewer] Sent final result to s5825_final
```

3.3. Poison message:
Created the dead-letter topic s5825_dead_letter using
docker exec hw8-kafka kafka-topics --create. The worker is set up so that any message with trialTitle POISON_TEST_TRIGGER_CRASH always fails, which simulates a permanently broken message.

```
rutujasaste@Rutujas-MacBook-Pro data260-5825 %  source /Users/rutujasaste/Documents/data260-5825/venv/bin/activate
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % docker exec hw8-kafka kafka-topics --create --topic s5825_dead_letter --partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid issues it is best to use either, but not both.
Created topic s5825_dead_letter.
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 %
```

After rebuilding the docker with the code of handling this poison message

```
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % docker compose up -d --build worker
docker logs hw8-worker
WARN[0000] /Users/rutujasaste/Documents/data260-5825/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 1.1s (12/12) FINISHED
 => [internal] load local bake definitions
 => => reading from stdin 574B
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 296B
 => [internal] load metadata for docker.io/library/python:3.11-slim
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/python:3.11-slim@sha256:673719e071ecf27434fd2635bbd8d00f49f4fdd81f243143438a42ced8bedf60
 => => resolve docker.io/library/python:3.11-slim@sha256:673719e071ecf27434fd2635bbd8d00f49f4fdd81f243143438a42ced8bedf60
 => [internal] load build context
 => => transferring context: 7.31kB
 => CACHED [2/5] WORKDIR /app
 => CACHED [3/5] COPY requirements.txt .
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
 => [5/5] COPY . .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:4d0c08137c9bca3821b370a94369df84fc78c771a0adf0e354e7a828f5cd3a13
 => => exporting config sha256:9d1642098223212818cbbd4e0beecdf32a763457d70b79559efde181ff7ecc4e
 => => exporting attestation manifest sha256:c07965363734f60633c0f088372dd03721ea457859fc6a1dc9aef16e8f8d26ff
 => => exporting manifest list sha256:d15a6b60efc2ecc44488ec0a933d9a240da83553e71ba4a5584046bb47bd517f
 => => naming to docker.io/library/data260-5825-worker:latest
 => => unpacking to docker.io/library/data260-5825-worker:latest
 => resolving provenance for metadata file
[+] up 4/4
 ✔ Image data260-5825-worker Built
 ✔ Container hw8-zookeeper   Running
 ✔ Container hw8-kafka       Running
 ✔ Container hw8-worker      Started
/app/worker.py:104: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
/app/worker.py:174: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
Consumer is ready and listening...
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 %
```

Code:

```python
MAX_RETRIES = 3

# Deliberate poison-message trigger for the Part 3.III demonstration:
# a message with this exact sentinel title always fails, simulating a
# permanently broken/corrupted message (not a normal validation error).
if isinstance(data, dict) and data.get("trialTitle") == "POISON_TEST_TRIGGER_CRASH":
    raise RuntimeError("Simulated processing crash (poison message test)")
```

```python
try:
    response = process_operation(correlation_id, operation, data)
    processed_requests[correlation_id] = response

    reply_to = request.get("reply_to", RESPONSE_TOPIC)
    producer.send(reply_to, value=response)
    producer.flush()
    print(f"[worker] Sent response to {reply_to}: {response}")

except Exception as e:
    new_retry_count = retry_count + 1
    print(f"[worker] ERROR processing correlation_id={correlation_id} "
          f"(attempt {new_retry_count}/{MAX_RETRIES}): {e}")

    if new_retry_count <= MAX_RETRIES:
        retried_request = dict(request)
        retried_request["retry_count"] = new_retry_count
        producer.send(REQUEST_TOPIC, value=retried_request)
        producer.flush()
        print(f"[worker] Requeued correlation_id={correlation_id} for retry "
              f"{new_retry_count}/{MAX_RETRIES} (other messages can be processed "
              f"in between - this does not block the queue)")
    else:
        dead_letter_message = {
            "correlation_id": correlation_id,
            "original_request": request,
            "error": str(e),
            "failed_attempts": new_retry_count,
        }
        producer.send(DEAD_LETTER_TOPIC, value=dead_letter_message)
        producer.flush()
        print(f"[worker] GIVING UP after {new_retry_count} attempts - sent "
              f"correlation_id={correlation_id} to {DEAD_LETTER_TOPIC}")
```

A malformed message sandwiched between two normal requests. Both normal requests succeeded, so the poison message did not block the queue. The poison message was retried 3 times, then moved to the dead-letter topic

```
rutujasaste@Rutujas-MacBook-Pro data260-5825 %  source /Users/rutujasaste/Documents/data260-5825/venv/bin/activate
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % cd ~/Documents/data260-5825/code/messaging/part3
source ../../../venv/bin/activate
python test_poison_message.py
/Users/rutujasaste/Documents/data260-5825/code/messaging/part3/test_poison_message.py:125: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
/Users/rutujasaste/Documents/data260-5825/code/messaging/part3/test_poison_message.py:61: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  response_consumer = KafkaConsumer(
/Users/rutujasaste/Documents/data260-5825/code/messaging/part3/test_poison_message.py:70: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  dead_letter_consumer = KafkaConsumer(
[test] normal_id_1 = 36848a2e-5add-454a-9400-13d57711424d
[test] poison_id   = 17f042b7-2f2a-4404-af7f-3dbbe7d0f4e5
[test] normal_id_2 = c62429b0-9b73-4d41-abd9-127e431d5596

[test] Sending normal request #1 ...
[test] Sending POISON request (trialTitle=POISON_TEST_TRIGGER_CRASH) ...
[test] Sending normal request #2 (right after the poison message) ...

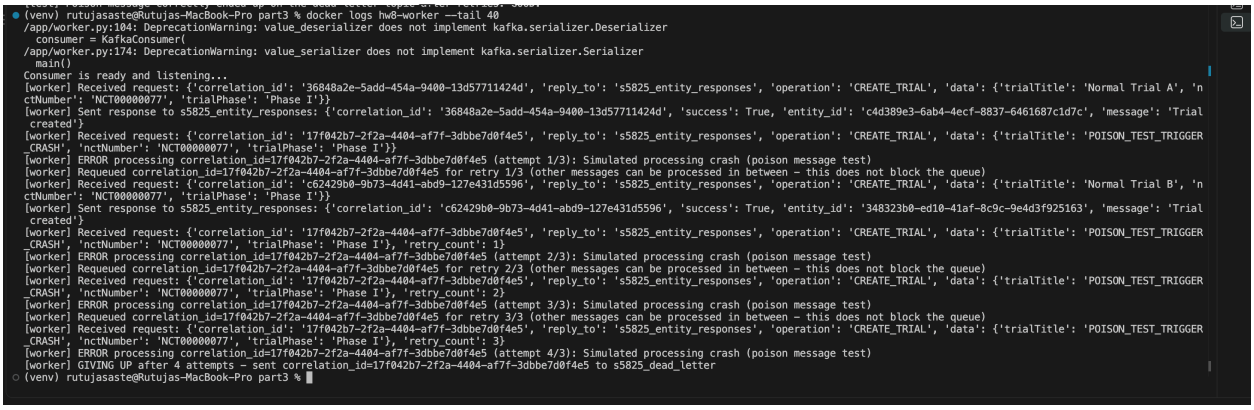
[test] Waiting for responses to the two NORMAL requests ...
[test] Normal request #1 response: {'correlation_id': '36848a2e-5add-454a-9400-13d57711424d', 'success': True, 'entity_id': 'c4d389e3-6ab4-4ecf-8837-6461687c1d7c', 'message': 'Trial created'}
[test] Normal request #2 response: {'correlation_id': 'c62429b0-9b73-4d41-abd9-127e431d5596', 'success': True, 'entity_id': '348323b0-ed10-41af-8c9c-9e4d3f925163', 'message': 'Trial created'}
[test] BOTH normal requests succeeded -> poison message did NOT block later messages. GOOD.

[test] Waiting for the poison message to appear on the dead-letter topic (it must retry and fail a bounded number of times first, so this takes a few seconds) ...

[test] Dead-letter result for poison_id: {'correlation_id': '17f042b7-2f2a-4404-af7f-3dbbe7d0f4e5', 'original_request': {'correlation_id': '17f042b7-2f2a-4404-af7f-3dbbe7d0f4e5', 'reply_to': 's582
5_entity_responses', 'operation': 'CREATE_TRIAL', 'data': {'trialTitle': 'POISON_TEST_TRIGGER_CRASH', 'nctNumber': 'NCT00000077', 'trialPhase': 'Phase I'}, 'retry_count': 3}, 'error': 'Simulated p
rocessing crash (poison message test)', 'failed_attempts': 4}
[test] Poison message correctly ended up on the dead-letter topic after retries. GOOD.
```

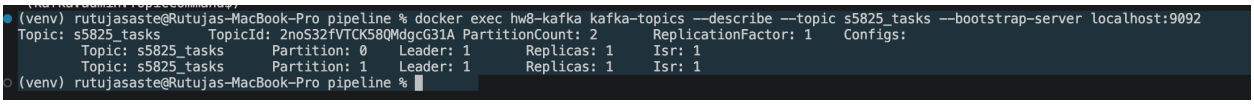
Worker Retry and Dead-Letter Log:
Here Normal Trial A processed and answered.
The poison message fails (attempt 1/3) and is requeued.
Normal Trial B is processed in between, which shows the queue wasn't blocked.

The poison message fails and is requeued at retry 2/3 and 3/3.
Attempt 4/3 fails and the worker prints
`[worker] GIVING UP after 4 attempts - sent correlation_id=17f042b7-2f2a-4404-af7f-3dbbe7d0f4e5 to s5825_dead_letter`

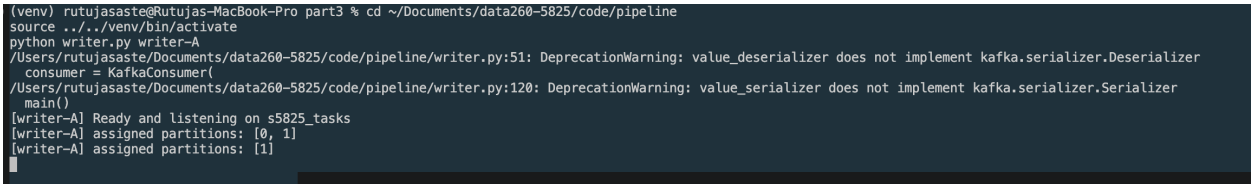


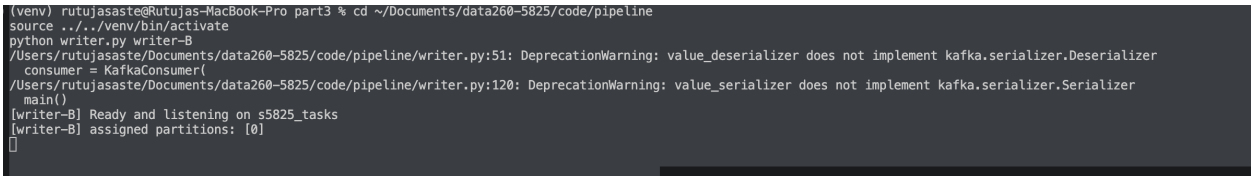
4. Ordering:

s5825_tasks has 2 partitions, Partition 0 and Partition 1: Hence two Writer instances in one consumer group can each own one partition.

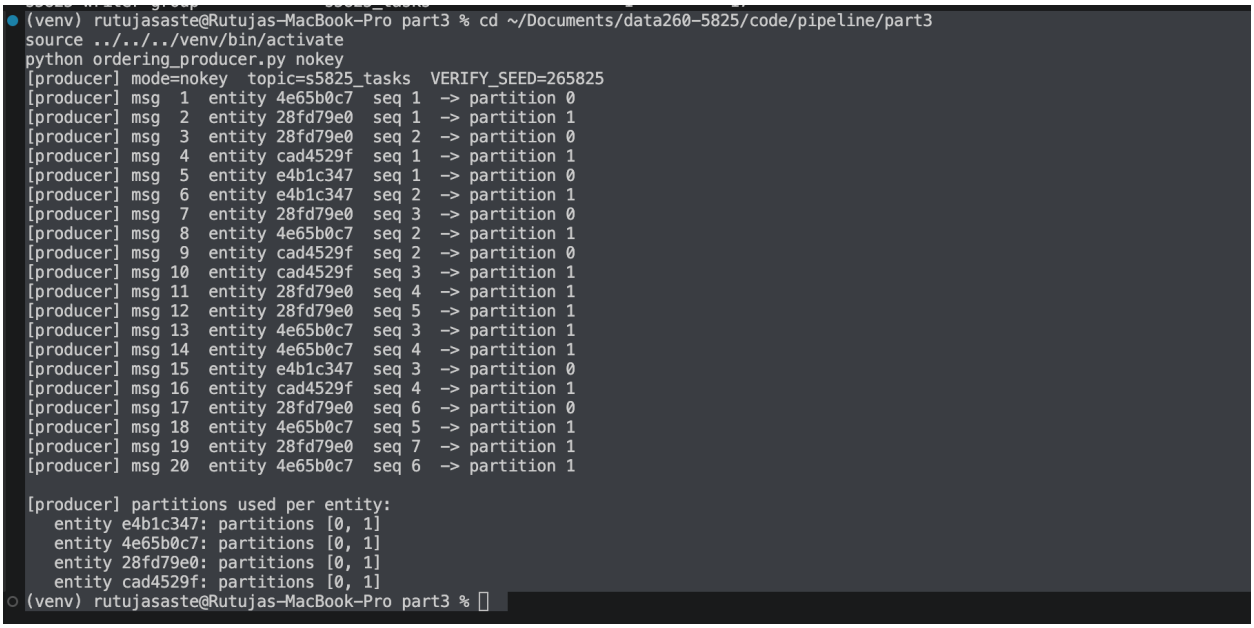


Two Writers in One Consumer Group:
Writer A and Writer B share one consumer group on s5825_tasks. Kafka split the 2 partitions between them: Writer A owns partition 1, Writer B owns partition 0.

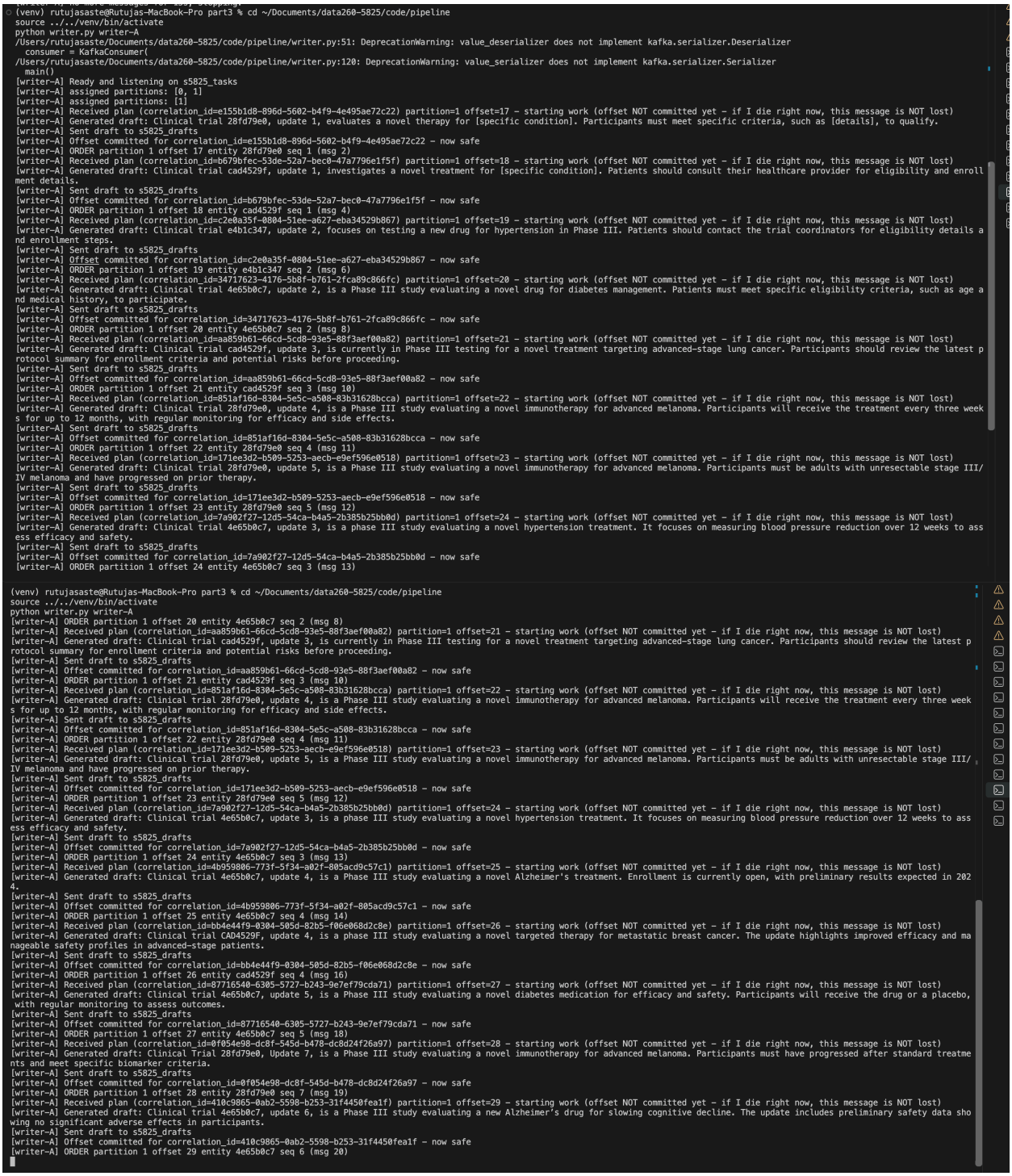




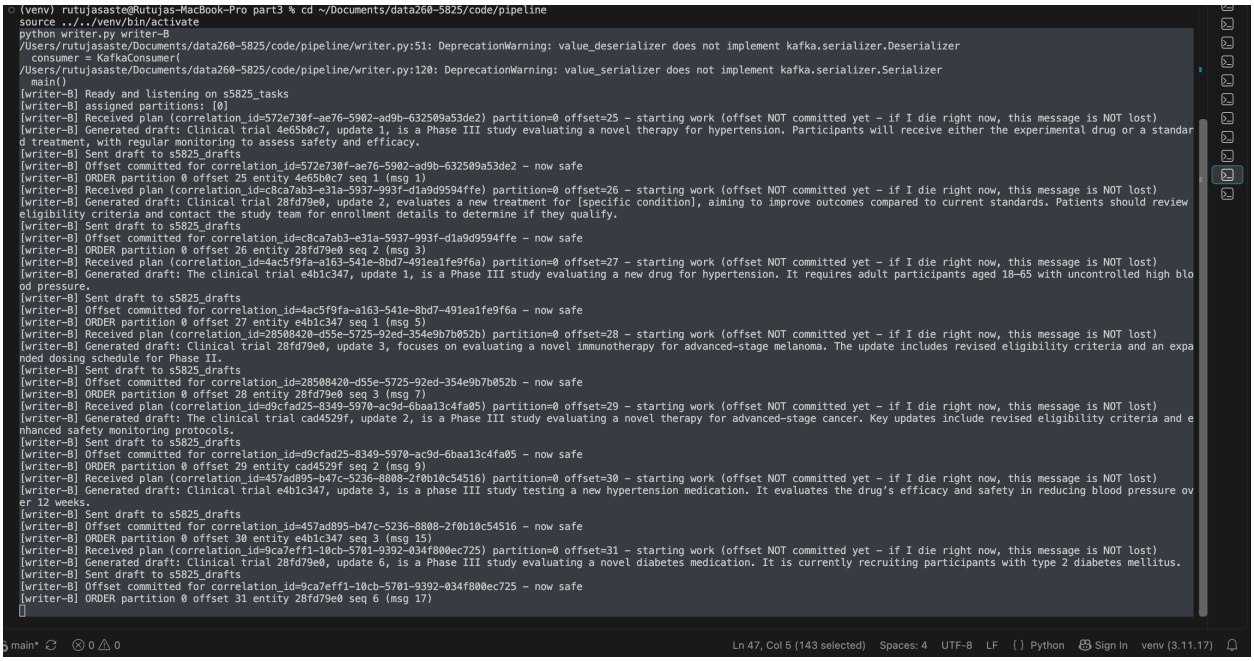
20 related messages about 4 trials (VERIFY_SEED = 265825) sent with no message key. Each trial's messages were spread across both partitions (partitions [0, 1] for every entity), so Kafka cannot keep them in order.



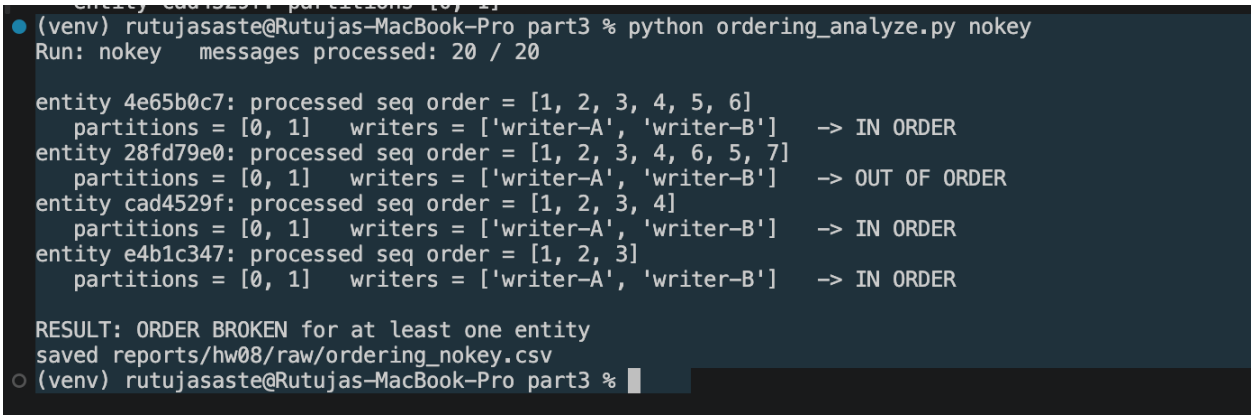
Run 1: With no key (Writer's processing order)
Two real Writers (writer.py) consuming in parallel, Writer A from partition 1 and Writer B from partition 0. Messages for the same trial, such as 28fd79e0 and 4e65b0c7, were handled by both Writers, so their processing order is mixed. Each Writer commits its offset only after finishing a message.


(venv) rutujasaste@Rutujas-MacBook-Pro part3 % cd ~/Documents/data260-5825/code/pipeline
source ../../venv/bin/activate
python writer.py writer-A
/Users/rutujasaste/Documents/data260-5825/code/pipeline/writer.py:51: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
consumer = KafkaConsumer(
/Users/rutujasaste/Documents/data260-5825/code/pipeline/writer.py:120: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
main()
[writer-A] Ready and listening on s5825_tasks
[writer-A] assigned partitions: [0, 1]
[writer-A] assigned partitions: [1]
[writer-A] Received plan (correlation_id=e155b1d8-896d-5602-b4f9-4e495ae72c22) partition=1 offset=17 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial 28fd79e0, update 1, evaluates a novel therapy for [specific condition]. Participants must meet specific criteria, such as [details], to qualify.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=e155b1d8-896d-5602-b4f9-4e495ae72c22 - now safe
[writer-A] ORDER partition 1 offset 17 entity 28fd79e0 seq 1 (msg 2)
[writer-A] Received plan (correlation_id=b679bfec-53de-52a7-bec0-47a7796e1f5f) partition=1 offset=18 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial cad4529f, update 1, investigates a novel treatment for [specific condition]. Patients should consult their healthcare provider for eligibility and enrollment details.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=b679bfec-53de-52a7-bec0-47a7796e1f5f - now safe
[writer-A] ORDER partition 1 offset 18 entity cad4529f seq 1 (msg 4)
[writer-A] Received plan (correlation_id=c2e0a35f-0804-51ee-a627-eba34529b867) partition=1 offset=19 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial e4b1c347, update 2, focuses on testing a new drug for hypertension in Phase III. Patients should contact the trial coordinators for eligibility details and enrollment steps.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=c2e0a35f-0804-51ee-a627-eba34529b867 - now safe
[writer-A] ORDER partition 1 offset 19 entity e4b1c347 seq 2 (msg 6)
[writer-A] Received plan (correlation_id=34717623-4176-5b8f-b761-2fca89c866fc) partition=1 offset=20 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial 4e65b0c7, update 2, is a Phase III study evaluating a novel drug for diabetes management. Patients must meet specific eligibility criteria, such as age and medical history, to participate.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=34717623-4176-5b8f-b761-2fca89c866fc - now safe
[writer-A] ORDER partition 1 offset 20 entity 4e65b0c7 seq 2 (msg 8)
[writer-A] Received plan (correlation_id=aa859b61-66cd-5cd8-93e5-88f3aef00a82) partition=1 offset=21 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial cad4529f, update 3, is currently in Phase III testing for a novel treatment targeting advanced-stage lung cancer. Participants should review the latest protocol summary for enrollment criteria and potential risks before proceeding.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=aa859b61-66cd-5cd8-93e5-88f3aef00a82 - now safe
[writer-A] ORDER partition 1 offset 21 entity cad4529f seq 3 (msg 10)
[writer-A] Received plan (correlation_id=851af16d-8304-5e5c-a508-83b31628bcca) partition=1 offset=22 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial 28fd79e0, update 4, is a Phase III study evaluating a novel immunotherapy for advanced melanoma. Participants will receive the treatment every three weeks for up to 12 months, with regular monitoring for efficacy and side effects.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=851af16d-8304-5e5c-a508-83b31628bcca - now safe
[writer-A] ORDER partition 1 offset 22 entity 28fd79e0 seq 4 (msg 11)
[writer-A] Received plan (correlation_id=171ee3d2-b509-5253-aecb-e9ef596e0518) partition=1 offset=23 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial 28fd79e0, update 5, is a Phase III study evaluating a novel immunotherapy for advanced melanoma. Participants must be adults with unresectable stage III/IV melanoma and have progressed on prior therapy.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=171ee3d2-b509-5253-aecb-e9ef596e0518 - now safe
[writer-A] ORDER partition 1 offset 23 entity 28fd79e0 seq 5 (msg 12)
[writer-A] Received plan (correlation_id=7a902f27-12d5-54ca-b4a5-2b385b25bb0d) partition=1 offset=24 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial 4e65b0c7, update 3, is a phase III study evaluating a novel hypertension treatment. It focuses on measuring blood pressure reduction over 12 weeks to assess efficacy and safety.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=7a902f27-12d5-54ca-b4a5-2b385b25bb0d - now safe
[writer-A] ORDER partition 1 offset 24 entity 4e65b0c7 seq 3 (msg 13)
(venv) rutujasaste@Rutujas-MacBook-Pro part3 % cd ~/Documents/data260-5825/code/pipeline
source ../../venv/bin/activate
python writer.py writer-A
[writer-A] ORDER partition 1 offset 20 entity 4e65b0c7 seq 2 (msg 8)
[writer-A] Received plan (correlation_id=aa859b61-66cd-5cd8-93e5-88f3aef00a82) partition=1 offset=21 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial cad4529f, update 3, is currently in Phase III testing for a novel treatment targeting advanced-stage lung cancer. Participants should review the latest protocol summary for enrollment criteria and potential risks before proceeding.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=aa859b61-66cd-5cd8-93e5-88f3aef00a82 - now safe
[writer-A] ORDER partition 1 offset 21 entity cad4529f seq 3 (msg 10)
[writer-A] Received plan (correlation_id=851af16d-8304-5e5c-a508-83b31628bcca) partition=1 offset=22 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial 28fd79e0, update 4, is a Phase III study evaluating a novel immunotherapy for advanced melanoma. Participants will receive the treatment every three weeks for up to 12 months, with regular monitoring for efficacy and side effects.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=851af16d-8304-5e5c-a508-83b31628bcca - now safe
[writer-A] ORDER partition 1 offset 22 entity 28fd79e0 seq 4 (msg 11)
[writer-A] Received plan (correlation_id=171ee3d2-b509-5253-aecb-e9ef596e0518) partition=1 offset=23 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial 28fd79e0, update 5, is a Phase III study evaluating a novel immunotherapy for advanced melanoma. Participants must be adults with unresectable stage III/IV melanoma and have progressed on prior therapy.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=171ee3d2-b509-5253-aecb-e9ef596e0518 - now safe
[writer-A] ORDER partition 1 offset 23 entity 28fd79e0 seq 5 (msg 12)
[writer-A] Received plan (correlation_id=7a902f27-12d5-54ca-b4a5-2b385b25bb0d) partition=1 offset=24 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial 4e65b0c7, update 3, is a phase III study evaluating a novel hypertension treatment. It focuses on measuring blood pressure reduction over 12 weeks to assess efficacy and safety.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=7a902f27-12d5-54ca-b4a5-2b385b25bb0d - now safe
[writer-A] ORDER partition 1 offset 24 entity 4e65b0c7 seq 3 (msg 13)
[writer-A] Received plan (correlation_id=4b959806-773f-5f34-a02f-805acd9c57c1) partition=1 offset=25 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial 4e65b0c7, update 4, is a Phase III study evaluating a novel Alzheimer's treatment. Enrollment is currently open, with preliminary results expected in 2024.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=4b959806-773f-5f34-a02f-805acd9c57c1 - now safe
[writer-A] ORDER partition 1 offset 25 entity 4e65b0c7 seq 4 (msg 14)
[writer-A] Received plan (correlation_id=bb4e44f9-0304-505d-82b5-f06e068d2c8e) partition=1 offset=26 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial CAD4529F, update 4, is a phase III study evaluating a novel targeted therapy for metastatic breast cancer. The update highlights improved efficacy and manageable safety profiles in advanced-stage patients.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=bb4e44f9-0304-505d-82b5-f06e068d2c8e - now safe
[writer-A] ORDER partition 1 offset 26 entity cad4529f seq 4 (msg 16)
[writer-A] Received plan (correlation_id=87716540-6305-5727-b243-9e7ef79cda71) partition=1 offset=27 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial 4e65b0c7, update 5, is a Phase III study evaluating a novel diabetes medication for efficacy and safety. Participants will receive the drug or a placebo, with regular monitoring to assess outcomes.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=87716540-6305-5727-b243-9e7ef79cda71 - now safe
[writer-A] ORDER partition 1 offset 27 entity 4e65b0c7 seq 5 (msg 18)
[writer-A] Received plan (correlation_id=0f054e98-dc8f-545d-b478-dc8d24f26a97) partition=1 offset=28 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical Trial 28fd79e0, Update 7, is a Phase III study evaluating a novel immunotherapy for advanced melanoma. Participants must have progressed after standard treatments and meet specific biomarker criteria.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=0f054e98-dc8f-545d-b478-dc8d24f26a97 - now safe
[writer-A] ORDER partition 1 offset 28 entity 28fd79e0 seq 7 (msg 19)
[writer-A] Received plan (correlation_id=410c9865-0ab2-5598-b253-31f4450fea1f) partition=1 offset=29 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial 4e65b0c7, update 6, is a Phase III study evaluating a new Alzheimer's drug for slowing cognitive decline. The update includes preliminary safety data showing no significant adverse effects in participants.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=410c9865-0ab2-5598-b253-31f4450fea1f - now safe
[writer-A] ORDER partition 1 offset 29 entity 4e65b0c7 seq 6 (msg 20)

Writer B

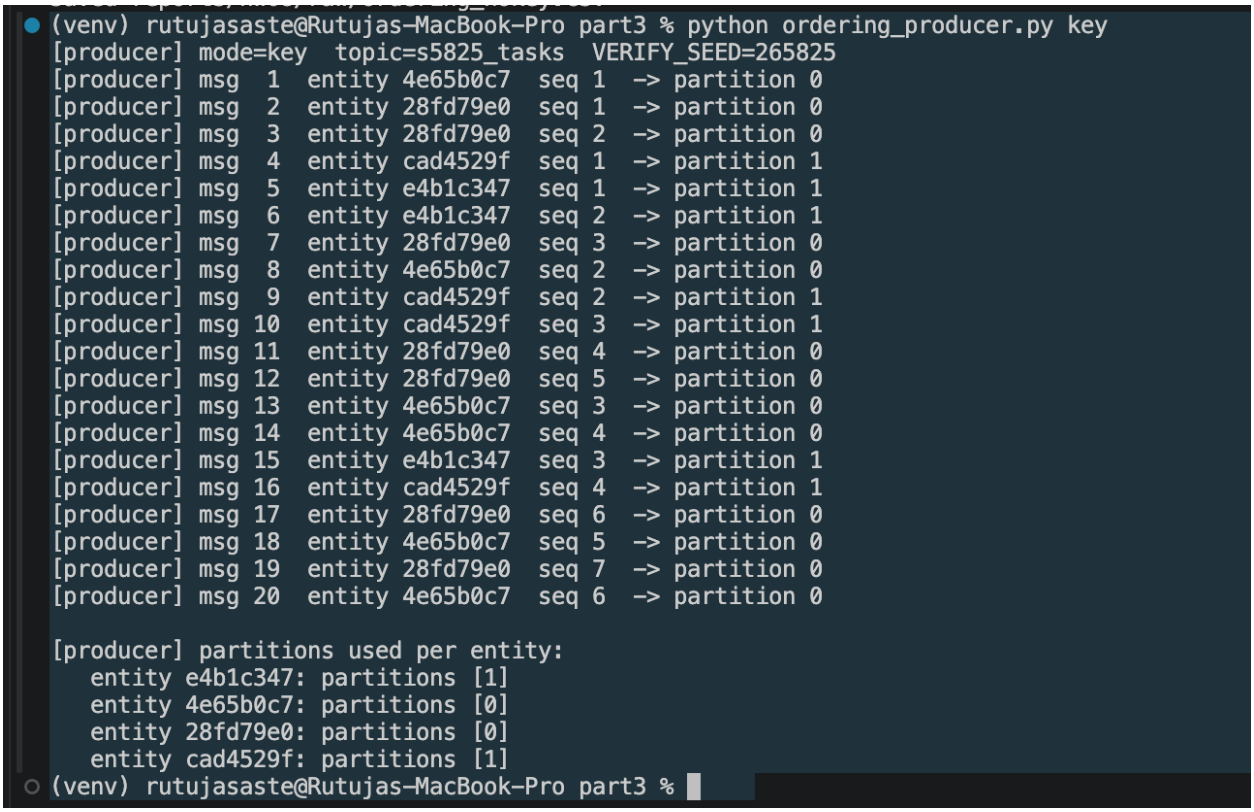


Result:
Output without a message key, related messages were spread across both partitions and handled by two Writers. Trial 28fd79e0 was processed out of order (update 6 before update 5). The other trials happened to finish in order, but nothing guaranteed it.



Run 2: With key (same setup as Run 1, with two Writers in one consumer group, one partition each)

The same 20 messages (VERIFY_SEED = 265825) sent with message key = entity_id. Every trial now lands in exactly one partition, so Kafka keeps its messages together in order.



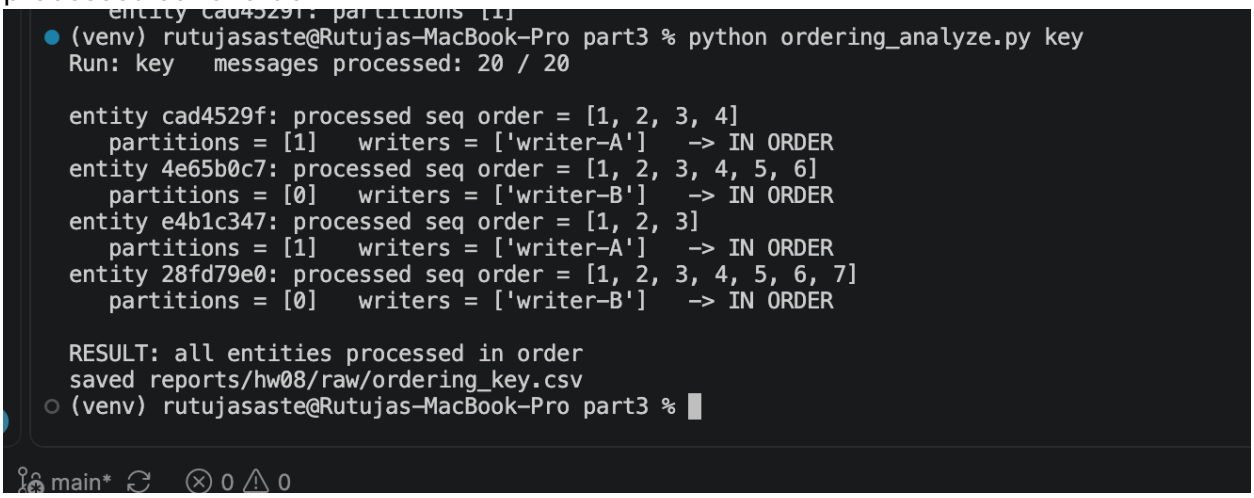
Writers' Processing Order - Writer A offsets 30 to 36, and Writer B offsets 32 to 44)

With key = entity_id, each trial's messages sit in one partition and are processed by one Writer in sequence. Writer B handled trials 4e65b0c7 and 28fd79e0 (partition 0), and Writer A handled cad4529f and e4b1c347 (partition 1).

```
(venv) rutujasaste@Rutujas-MacBook-Pro part3 % cd ~/Documents/data260-5825/code/pipeline
source ../../venv/bin/activate
python writer.py writer-A
[writer-A] ORDER partition 1 offset 27 entity 4e65b0c7 seq 5 (msg 18)
[writer-A] Received plan (correlation_id=0f054e98-dc8f-545d-b478-dc8d24f26a97) partition=1 offset=28 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical Trial 28fd79e0, Update 7, is a Phase III study evaluating a novel immunotherapy for advanced melanoma. Participants must have progressed after standard treatme
nts and meet specific biomarker criteria.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=0f054e98-dc8f-545d-b478-dc8d24f26a97 - now safe
[writer-A] ORDER partition 1 offset 28 entity 28fd79e0 seq 7 (msg 19)
[writer-A] Received plan (correlation_id=410c9865-0ab2-5598-b253-31f4450fea1f) partition=1 offset=29 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial 4e65b0c7, update 6, is a Phase III study evaluating a new Alzheimer's drug for slowing cognitive decline. The update includes preliminary safety data sho
wing no significant adverse effects in participants.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=410c9865-0ab2-5598-b253-31f4450fea1f - now safe
[writer-A] ORDER partition 1 offset 29 entity 4e65b0c7 seq 6 (msg 20)
[writer-A] Received plan (correlation_id=7356a628-3c4c-5f02-a540-4413c0b50e50) partition=1 offset=30 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial cad4529f, update 1, evaluates a novel therapy for [specific condition], currently in phase III. Patients should consult the trial's official documentatio
n or healthcare providers for detailed eligibility criteria and potential risks.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=7356a628-3c4c-5f02-a540-4413c0b50e50 - now safe
[writer-A] ORDER partition 1 offset 30 entity cad4529f seq 1 (msg 4)
[writer-A] Received plan (correlation_id=77fd5904-65ee-52f9-b619-466bed98bbfa) partition=1 offset=31 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial e4b1c347, update 1, is a Phase III study evaluating a novel immunotherapy for advanced melanoma. Patients should contact the research team for eligibilit
y details and enrollment steps.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=77fd5904-65ee-52f9-b619-466bed98bbfa - now safe
[writer-A] ORDER partition 1 offset 31 entity e4b1c347 seq 1 (msg 5)
[writer-A] Received plan (correlation_id=889f12e0-813b-57d0-ad7a-fc35749c5785) partition=1 offset=32 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial e4b1c347, update 2, focuses on evaluating a novel treatment for a specific condition. Patients should consult the trial's official documentation or their
 healthcare provider for eligibility criteria and potential risks.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=889f12e0-813b-57d0-ad7a-fc35749c5785 - now safe
[writer-A] ORDER partition 1 offset 32 entity e4b1c347 seq 2 (msg 6)
[writer-A] Received plan (correlation_id=5aa9a6ef-4b35-55cd-86d2-0c86916b9847) partition=1 offset=33 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial cad4529f, update 2, is a Phase III study evaluating the efficacy and safety of a novel targeted therapy for advanced non-small cell lung cancer. The tria
l's primary endpoint is progression-free survival, with results expected in early 2024 to guide future treatment standards.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=5aa9a6ef-4b35-55cd-86d2-0c86916b9847 - now safe
[writer-A] ORDER partition 1 offset 33 entity cad4529f seq 2 (msg 9)
[writer-A] Received plan (correlation_id=c24b3506-e5d2-51ac-82c6-b5a253a31620) partition=1 offset=34 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical trial cad4529f, update 3, focuses on evaluating a new treatment for [specific condition]. Patients should consult their healthcare provider to determine eligib
ility and understand potential risks and benefits before participating.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=c24b3506-e5d2-51ac-82c6-b5a253a31620 - now safe
[writer-A] ORDER partition 1 offset 34 entity cad4529f seq 3 (msg 10)
[writer-A] Received plan (correlation_id=d5d7fbd4-8db6-5d3f-a876-9acd768e0d59) partition=1 offset=35 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: Clinical Trial e4b1c347, Update 3, investigates a novel treatment for [specific condition], currently in Phase II. Patients should contact the study team for eligibilit
y details and enrollment steps.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=d5d7fbd4-8db6-5d3f-a876-9acd768e0d59 - now safe
[writer-A] ORDER partition 1 offset 35 entity e4b1c347 seq 3 (msg 15)
[writer-A] Received plan (correlation_id=aa5f4302-abaf-58dc-a39f-1636f07fafc3) partition=1 offset=36 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-A] Generated draft: The clinical trial cad4529f, update 4, is currently recruiting participants to evaluate a new treatment for a specific condition. Patients should consult the trial's of
ficial documentation or their healthcare provider for detailed eligibility criteria and potential risks.
[writer-A] Sent draft to s5825_drafts
[writer-A] Offset committed for correlation_id=aa5f4302-abaf-58dc-a39f-1636f07fafc3 - now safe
[writer-A] ORDER partition 1 offset 36 entity cad4529f seq 4 (msg 16)
```

```
(venv) rutujasaste@Rutujas-MacBook-Pro part3 % cd ~/Documents/data260-5825/code/pipeline
source ../../venv/bin/activate
python writer.py writer-B
[writer-B] Offset committed for correlation_id=77975620-0f44-592d-b74a-b1bd4e41410b - now safe
[writer-B] ORDER partition 0 offset 35 entity 28fd79e0 seq 3 (msg 7)
[writer-B] Received plan (correlation_id=941f808a-ec27-5c6e-9976-4f4c847c801c) partition=0 offset=36 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-B] Generated draft: Clinical trial 4e65b0c7, update 2, is a Phase III study testing a new Alzheimer's drug for slowing cognitive decline. Patients must be 65+ with early-stage Alzheimer's
and meet specific biomarker criteria to qualify.
[writer-B] Sent draft to s5825_drafts
[writer-B] Offset committed for correlation_id=941f808a-ec27-5c6e-9976-4f4c847c801c - now safe
[writer-B] ORDER partition 0 offset 36 entity 4e65b0c7 seq 2 (msg 8)
[writer-B] Received plan (correlation_id=da56c013-6389-5d90-a34e-ebd68ec5fcbb) partition=0 offset=37 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-B] Generated draft: Clinical trial 28fd79e0, update 4, is a Phase III study evaluating a novel drug for hypertension. It focuses on assessing efficacy and safety in adults with uncontrolle
d high blood pressure.
[writer-B] Sent draft to s5825_drafts
[writer-B] Offset committed for correlation_id=da56c013-6389-5d90-a34e-ebd68ec5fcbb - now safe
[writer-B] ORDER partition 0 offset 37 entity 28fd79e0 seq 4 (msg 11)
[writer-B] Received plan (correlation_id=1009f037-f847-521b-b7e9-cbdc8f1ff498) partition=0 offset=38 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-B] Generated draft: Clinical trial 28fd79e0, update 5, is currently in Phase 3 testing for a novel hypertension treatment. Participants must meet specific eligibility criteria, including b
lood pressure thresholds and prior treatment history.
[writer-B] Sent draft to s5825_drafts
[writer-B] Offset committed for correlation_id=1009f037-f847-521b-b7e9-cbdc8f1ff498 - now safe
[writer-B] ORDER partition 0 offset 38 entity 28fd79e0 seq 5 (msg 12)
[writer-B] Received plan (correlation_id=c62f8a26-a9ef-5a4c-aaa6-c8017f80ca37) partition=0 offset=39 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-B] Generated draft: Clinical trial 4e65b0c7, update 3, is a phase III study evaluating a new cancer drug's efficacy. The latest results show a 30% improvement in progression-free survival
compared to standard treatment.
[writer-B] Sent draft to s5825_drafts
[writer-B] Offset committed for correlation_id=c62f8a26-a9ef-5a4c-aaa6-c8017f80ca37 - now safe
[writer-B] ORDER partition 0 offset 39 entity 4e65b0c7 seq 3 (msg 13)
[writer-B] Received plan (correlation_id=c04c9ac3-c4ff-58b8-a364-dbdc92a93901) partition=0 offset=40 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-B] Generated draft: Clinical trial 4e65b0c7, update 4, is a Phase III study evaluating a novel Alzheimer's treatment. Participants must be early-stage patients aged 55-85 with mild cogniti
ve impairment.
[writer-B] Sent draft to s5825_drafts
[writer-B] Offset committed for correlation_id=c04c9ac3-c4ff-58b8-a364-dbdc92a93901 - now safe
[writer-B] ORDER partition 0 offset 40 entity 4e65b0c7 seq 4 (msg 14)
[writer-B] Received plan (correlation_id=2d8fd7f9-4277-5423-8715-dca8d060874e) partition=0 offset=41 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-B] Generated draft: Clinical trial 28fd79e0, update 6, is a Phase III study evaluating a new diabetes drug's efficacy and safety. Patients should contact the trial team for eligibility det
ails and enrollment steps.
[writer-B] Sent draft to s5825_drafts
[writer-B] Offset committed for correlation_id=2d8fd7f9-4277-5423-8715-dca8d060874e - now safe
[writer-B] ORDER partition 0 offset 41 entity 28fd79e0 seq 6 (msg 17)
[writer-B] Received plan (correlation_id=48e69b74-b492-5ee8-bedc-513dcb3c802c) partition=0 offset=42 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-B] Generated draft: Clinical trial 4e65b0c7, update 5, is a Phase III study evaluating a novel immunotherapy for advanced melanoma. The trial now includes expanded eligibility for patients
 with BRAF mutations and updated safety data showing manageable side effects.
[writer-B] Sent draft to s5825_drafts
[writer-B] Offset committed for correlation_id=48e69b74-b492-5ee8-bedc-513dcb3c802c - now safe
[writer-B] ORDER partition 0 offset 42 entity 4e65b0c7 seq 5 (msg 18)
[writer-B] Received plan (correlation_id=e9102a27-4fc7-5980-a33b-22412b10f1bc) partition=0 offset=43 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-B] Generated draft: Clinical Trial 28fd79e0, Update 7, evaluates a new targeted therapy for advanced lung cancer. Participants may gain access to cutting-edge treatment while contributing
to medical research.
[writer-B] Sent draft to s5825_drafts
[writer-B] Offset committed for correlation_id=e9102a27-4fc7-5980-a33b-22412b10f1bc - now safe
[writer-B] ORDER partition 0 offset 43 entity 28fd79e0 seq 7 (msg 19)
[writer-B] Received plan (correlation_id=943e51c3-8e02-5a02-bd50-4926f317db4a) partition=0 offset=44 - starting work (offset NOT committed yet - if I die right now, this message is NOT lost)
[writer-B] Generated draft: Clinical trial 4e65b0c7, update 6, is a phase III study evaluating a novel Alzheimer's treatment. Enrollment is complete, with top-line results expected in Q1 2024.
[writer-B] Sent draft to s5825_drafts
[writer-B] Offset committed for correlation_id=943e51c3-8e02-5a02-bd50-4926f317db4a - now safe
[writer-B] ORDER partition 0 offset 44 entity 4e65b0c7 seq 6 (msg 20)
```

With message key = entity_id, all 20 messages were processed in order. Each trial stayed in one partition and was handled by one Writer. In Run 1 (no key), trial 28fd79e0 was processed out of order.



Summary of the results:

	Run 1: no key	Run 2: key = entity_id
Partitions used per trial	both [0, 1]	one only
Writers per trial	both A and B	one only
Trials out of order	1 of 4 (28fd79e0: 6 processed before 5)	0 of 4

Without a key, Kafka spreads messages across partitions, so two Writers can finish updates to the same trial in the wrong order. Using the trial's ID as the message key sends all of its messages to one partition, which only one Writer reads at a time, so their order is preserved. The 20 test messages were sent directly to s5825_tasks by a test script (ordering_producer.py); the Planner was not involved. Both Writers are instances of the real writer.py.

Partition-key code:

```
key = m["entity_id"] if mode == "key" else None   # <-- the partition key
md = producer.send(TASKS_TOPIC, key=key, value=m).get(timeout=10)
```

5. Comparison:

Before fix :Consumer Killed Mid-Work, Before Fix (Offset Committed on Receipt)
The Writer commits the offset the moment a message arrives, before doing the work. It was killed (Ctrl+C) during the LLM call, so no draft was ever produced for correlation_id 9ab8f7b8-30e6-4af3-abb7-88c8222aa691.

```
(venv) rutujasaste@Rutujas-MacBook-Pro pipeline % cd ~/Documents/data260-5825/code/pipeline/part3/before_fix
source ../../../../venv/bin/activate
python writer_before_fix.py
/Users/rutujasaste/Documents/data260-5825/code/pipeline/part3/before_fix/writer_before_fix.py:34: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
/Users/rutujasaste/Documents/data260-5825/code/pipeline/part3/before_fix/writer_before_fix.py:71: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
[writer-BEFORE-FIX] Ready and listening on s5825_tasks
```

```
(venv) rutujasaste@Rutujas-MacBook-Pro pipeline % cd ~/Documents/data260-5825/code/pipeline/part3/before_fix
source ../../../../venv/bin/activate
python writer_before_fix.py
/Users/rutujasaste/Documents/data260-5825/code/pipeline/part3/before_fix/writer_before_fix.py:34: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
/Users/rutujasaste/Documents/data260-5825/code/pipeline/part3/before_fix/writer_before_fix.py:71: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
[writer-BEFORE-FIX] Ready and listening on s5825_tasks
[writer-BEFORE-FIX] Received plan (correlation_id=9ab8f7b8-30e6-4af3-abb7-88c8222aa691) partition=1 offset=37
[writer-BEFORE-FIX] Offset committed IMMEDIATELY (before work) for correlation_id=9ab8f7b8-30e6-4af3-abb7-88c8222aa691
[writer-BEFORE-FIX] Starting LLM work ...
^CTraceback (most recent call last):
  File "/Users/rutujasaste/Documents/data260-5825/code/pipeline/part3/before_fix/writer_before_fix.py", line 71, in <module>
    main()
  File "/Users/rutujasaste/Documents/data260-5825/code/pipeline/part3/before_fix/writer_before_fix.py", line 60, in main
    draft = write_answer(task.get("question"), task.get("plan"))
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/rutujasaste/Documents/data260-5825/code/pipeline/part3/before_fix/writer_before_fix.py", line 30, in write_answer
    return client.complete(messages)
           ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/rutujasaste/Documents/data260-5825/code/pipeline/part3/before_fix/../../../../src/model_client.py", line 15, in complete
    response = self.llm.invoke(messages)
               ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_core/language_models/chat_models.py", line 488, in invoke
    self.generate_prompt(
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_core/language_models/chat_models.py", line 1877, in generate_prompt
    return self.generate(prompt_messages, stop=stop, callbacks=callbacks, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_core/language_models/chat_models.py", line 1684, in generate
    self._generate_with_cache(
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_core/language_models/chat_models.py", line 2022, in _generate_with_cache
    result = self._generate(
             ^^^^^^^^^^^^^^^
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_ollama/chat_models.py", line 1200, in _generate
    final_chunk = self._chat_stream_with_aggregation(
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_ollama/chat_models.py", line 1135, in _chat_stream_with_aggregation
    for chunk in self._iterate_over_stream(messages, stop, **kwargs):
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_ollama/chat_models.py", line 1224, in _iterate_over_stream
    for stream_resp in self._create_chat_stream(messages, stop, **kwargs):
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_ollama/chat_models.py", line 1122, in _create_chat_stream
    yield from self._client.chat(**chat_params)
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/ollama/_client.py", line 191, in inner
    for line in r.iter_lines():
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpx/_models.py", line 929, in iter_lines
    for text in self.iter_text():
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpx/_models.py", line 916, in iter_text
    for byte_content in self.iter_bytes():
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpx/_models.py", line 897, in iter_bytes
    for raw_bytes in self.iter_raw():
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpx/_models.py", line 951, in iter_raw
    for raw_stream_bytes in self.stream:
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpx/_client.py", line 153, in __iter__
    for chunk in self._stream:
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpx/_transports/default.py", line 127, in __iter__
```

Consumer Killed Mid-Work, Before Fix (Message Lost After Restart):
After restarting, the Writer is idle: the message 9ab8f7b8-30e6-4af3-abb7-88c8222aa691 is never redelivered, because its offset was already committed. The work is lost.

```
(venv) rutujasaste@Rutujas-MacBook-Pro before_fix % python writer_before_fix.py
/Users/rutujasaste/Documents/data260-5825/code/pipeline/part3/before_fix/writer_before_fix.py:34: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
/Users/rutujasaste/Documents/data260-5825/code/pipeline/part3/before_fix/writer_before_fix.py:71: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
[writer-BEFORE-FIX] Ready and listening on s5825_tasks
```

```
(venv) rutujasaste@Rutujas-MacBook-Pro pipeline % cd ~/Documents/data260-5825/code/pipeline/part3/before_fix
source ../../../../venv/bin/activate
python writer_before_fix.py
                           ^^^^^^^^^^^^^^^^^^^^^^^^

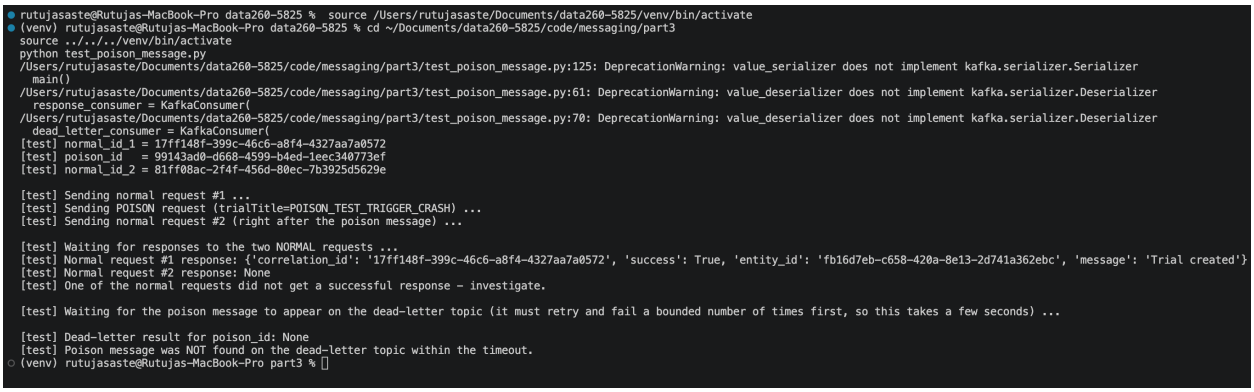
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_core/language_models/chat_models.py", line 488, in invoke
    self.generate_prompt(
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_core/language_models/chat_models.py", line 1877, in generate_prompt
    return self.generate(prompt_messages, stop=stop, callbacks=callbacks, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_core/language_models/chat_models.py", line 1684, in generate
    self._generate_with_cache(
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_core/language_models/chat_models.py", line 2022, in _generate_with_cache
    result = self._generate(
             ^^^^^^^^^^^^^^^
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_ollama/chat_models.py", line 1200, in _generate
    final_chunk = self._chat_stream_with_aggregation(
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_ollama/chat_models.py", line 1135, in _chat_stream_with_aggregation
    for chunk in self._iterate_over_stream(messages, stop, **kwargs):
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_ollama/chat_models.py", line 1224, in _iterate_over_stream
    for stream_resp in self._create_chat_stream(messages, stop, **kwargs):
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/langchain_ollama/chat_models.py", line 1122, in _create_chat_stream
    yield from self._client.chat(**chat_params)
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/ollama/_client.py", line 191, in inner
    for line in r.iter_lines():
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpx/_models.py", line 929, in iter_lines
    for text in self.iter_text():
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpx/_models.py", line 916, in iter_text
    for byte_content in self.iter_bytes():
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpx/_models.py", line 897, in iter_bytes
    for raw_bytes in self.iter_raw():
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpx/_models.py", line 951, in iter_raw
    for raw_stream_bytes in self.stream:
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpx/_client.py", line 153, in __iter__
    for chunk in self._stream:
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpx/_transports/default.py", line 127, in __iter__
    for part in self._httpcore_stream:
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpcore/_sync/connection_pool.py", line 407, in __iter__
    raise exc from None
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpcore/_sync/connection_pool.py", line 403, in __iter__
    for part in self._stream:
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpcore/_sync/http11.py", line 342, in __iter__
    raise exc
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpcore/_sync/http11.py", line 334, in __iter__
    for chunk in self._connection._receive_response_body(**kwargs):
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpcore/_sync/http11.py", line 203, in _receive_response_body
    event = self._receive_event(timeout=timeout)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpcore/_sync/http11.py", line 217, in _receive_event
    data = self._network_stream.read(
           ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/rutujasaste/Documents/data260-5825/venv/lib/python3.11/site-packages/httpcore/_backends/sync.py", line 128, in read
    return self._sock.recv(max_bytes)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^
KeyboardInterrupt

(venv) rutujasaste@Rutujas-MacBook-Pro before_fix %
```

Poison Message, Before Fix (Worker Crashes):
Without retries or a dead-letter topic, the worker processed the first normal message, then crashed on the poison message (RuntimeError, unhandled). The process is dead, so Normal Trial B behind it is never processed.

```
rutujasaste@Rutujas-MacBook-Pro data260-5825 %  source /Users/rutujasaste/Documents/data260-5825/venv/bin/activate
(venv) rutujasaste@Rutujas-MacBook-Pro data260-5825 % cd ~/Documents/data260-5825/code/messaging/part3/before_fix
source ../../../../venv/bin/activate
python worker_before_fix.py
/Users/rutujasaste/Documents/data260-5825/code/messaging/part3/before_fix/worker_before_fix.py:91: DeprecationWarning: value_deserializer does not implement kafka.serializer.Deserializer
  consumer = KafkaConsumer(
/Users/rutujasaste/Documents/data260-5825/code/messaging/part3/before_fix/worker_before_fix.py:137: DeprecationWarning: value_serializer does not implement kafka.serializer.Serializer
  main()
[worker-BEFORE-FIX] Consumer is ready and listening...
[worker] Received request: {'correlation_id': '17ff148f-399c-46c6-a8f4-4327aa7a0572', 'reply_to': 's5825_entity_responses', 'operation': 'CREATE_TRIAL', 'data': {'trialTitle': 'Normal Trial A'
, 'nctNumber': 'NCT00000077', 'trialPhase': 'Phase I'}}
[worker] Sent response to s5825_entity_responses: {'correlation_id': '17ff148f-399c-46c6-a8f4-4327aa7a0572', 'success': True, 'entity_id': 'fb16d7eb-c658-420a-8e13-2d741a362ebc', 'message': 'T
rial created'}
[worker] Received request: {'correlation_id': '99143ad0-d668-4599-b4ed-1eec340773ef', 'reply_to': 's5825_entity_responses', 'operation': 'CREATE_TRIAL', 'data': {'trialTitle': 'POISON_TEST_TRI
GGER_CRASH', 'nctNumber': 'NCT00000077', 'trialPhase': 'Phase I'}}
Traceback (most recent call last):
  File "/Users/rutujasaste/Documents/data260-5825/code/messaging/part3/before_fix/worker_before_fix.py", line 137, in <module>
    main()
  File "/Users/rutujasaste/Documents/data260-5825/code/messaging/part3/before_fix/worker_before_fix.py", line 127, in main
    response = process_operation(correlation_id, operation, data)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/rutujasaste/Documents/data260-5825/code/messaging/part3/before_fix/worker_before_fix.py", line 47, in process_operation
    raise RuntimeError("Simulated processing crash (poison message test)")
RuntimeError: Simulated processing crash (poison message test)
(venv) rutujasaste@Rutujas-MacBook-Pro before_fix %
```

Poison Message, Before Fix (Test Result):
Without retries or a dead-letter topic, the worker crashed on the poison message. Normal request #2, sent right after it, got no response, and the poison message never reached any dead-letter topic: one bad message blocked everything behind it.



Two-benefits/two-responsibilities comparison with the HW2 in-process graph:

In HW2, all the steps ran inside one process, so each step waited for the previous one to finish. Kafka improved this in two ways. First, if the Writer stopped while working, the message stayed safely in the s5825_tasks topic and was processed again when the Writer restarted. In HW2, a failure could stop or lose the whole workflow. Second, Kafka allowed two Writer instances to share the work and process messages in parallel, which was not possible with the single HW2 process. However, Kafka also created new responsibilities. Since Kafka can deliver the same message more than once, the worker must remember completed correlation_id values so it does not create duplicate trials. Also, messages may be processed out of order when multiple partitions are used, so I use entity_id as the message key to keep all messages for the same trial in one partition.

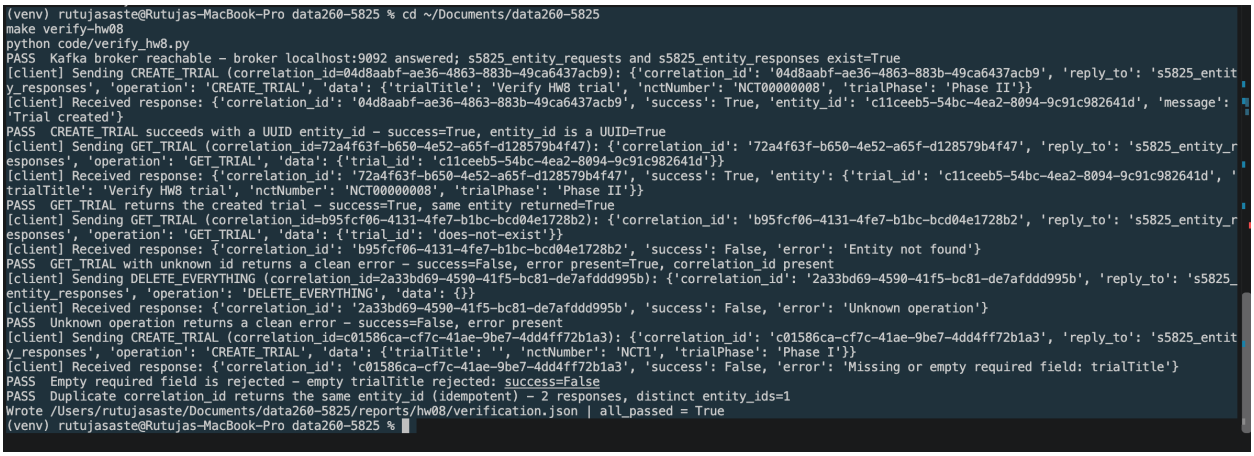
Failure Table:

Failure scenario	Before fix	After fix
Duplicate delivery	Same correlation_id ddb212a5-b506-410a-a16b-da238a05f255 replayed. Response #1 entity_id 8eca8d1b-b6c0-407c-a03f-0ebea33991e3, Response #2 entity_id 92826e87-2dc1-4fb2-9bc0-bf0d7c294121. "DIFFERENT entity_id each time -> NOT idempotent. A duplicate trial was created."	Same replay with correlation_id de829fe1-3b6e-4500-a15d-ed709f9e89f2 returned entity_id d21e877a-731b-4aa7-8042-9232803adf46 both times. Worker log: DUPLICATE detected for correlation_id=de829fe1-... - returning cached response, no new trial created
Consumer killed mid-work	Offset committed IMMEDIATELY (before work) for correlation_id=9ab8f7b8-30e6-4af3-abb7-88c8222aa691, then Writer killed during the LLM call. After restart, no Received plan: the message is lost.	Writer killed before the draft, restarted, same message 05038906-8b41-4514-bf2d-6418f0c6d409 redelivered and completed; Offset committed ... now safe printed only after the send.

Failure scenario	Before fix	After fix
Poison message	Worker crashed: RuntimeError: Simulated processing crash (poison message test). Normal request #2 response: None; poison not on any dead-letter topic.	GIVING UP after 4 attempts - sent correlation_id=17f042b7-2f2a-4404-af7f-3dbbe7d0f4e5 to s5825_dead_letter; both normal requests succeeded.
Two partitions/two consumers	Run 1, no key: trial 28fd79e0 processed as [1, 2, 3, 4, 6, 5, 7], OUT OF ORDER.	Run 2, key = entity_id: all four trials IN ORDER, each on one partition and one Writer.

“Before fix” for Consumer killed is a copy of the Writer that commits the offset immediately on receipt; “before fix” for Poison message is a copy of the worker without retry or dead-letter handling.

All 7 smoke-test checks pass:



AI_USE:

```
# AI_USE.md - HW8

1. What I used an AI assistant for, and what I did myself
I used AI to explain Kafka concepts (correlation IDs, consumer groups, offsets,
partition keys) in simple terms, to help implement and fix the Part 1 worker and also
the Part 2 Planner agent. I understood and tried to implement the Writer and reviewer
agents based on planner by understanding it first, started the Docker services,
collected the logs, and wrote the report.

2. An AI output that was wrong or unsuitable
For the Part 3 ordering test, the AI first suggested me to use a stand-in consumer way
of implementation instead of using my real Writer agent. That meant the ordering
```

evidence did not come from my actual system. I also nearly kept a report sentence that said Kafka topic names must avoid periods and underscores, which came from a Kafka warning message and was not a real rule.

3. How I detected the problem
I compared the work against the exact requirement, which asks for evidence from my own system, and saw that the stand-in did not match it. I checked the topic-name sentence against the actual warning in my terminal output.

4. What I changed and why it works now
I replaced the stand-in with two real Writer instances in one consumer group (writer.py with a name argument), and re-ran the no-key and key ordering runs so the data comes from the real pipeline. Now the ordering results (no key = out of order, entity_id key = in order) is what is expected.